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Humphrey et al.

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(54) **GAS INFUSER FOR LIQUIDS**

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2019, now Pat. No. 11,617,994.

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16, 2018, provisional application No. 62/628,035,
filed on Feb. 8, 2018.

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B01F 23/237 (2022.01)

B01F 101/14 (2022.01)

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(2022.01); **B01F 35/7172** (2022.01); **B01F**
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(2022.01); **B01F 23/23765** (2022.01); **B01F**
2101/14 (2022.01)

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23/23765; **B01F 25/30**; **B01F 35/7172**;
B01F 35/71805; **B01F 2101/14**

USPC **261/79.2**, **DIG. 7**
See application file for complete search history.

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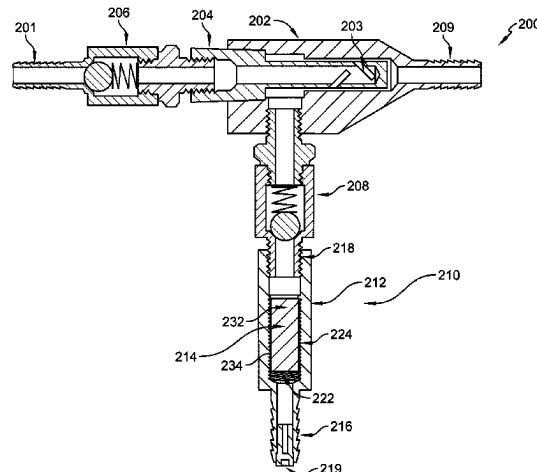
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LLP

(57) **ABSTRACT**

A gas infuser includes a mixing chamber and a liquid
injector. Gas is supplied into the mixing chamber and liquid
flows through openings of the injector to mix with the gas in
the mixing chamber and form a gas-infused liquid.

11 Claims, 17 Drawing Sheets



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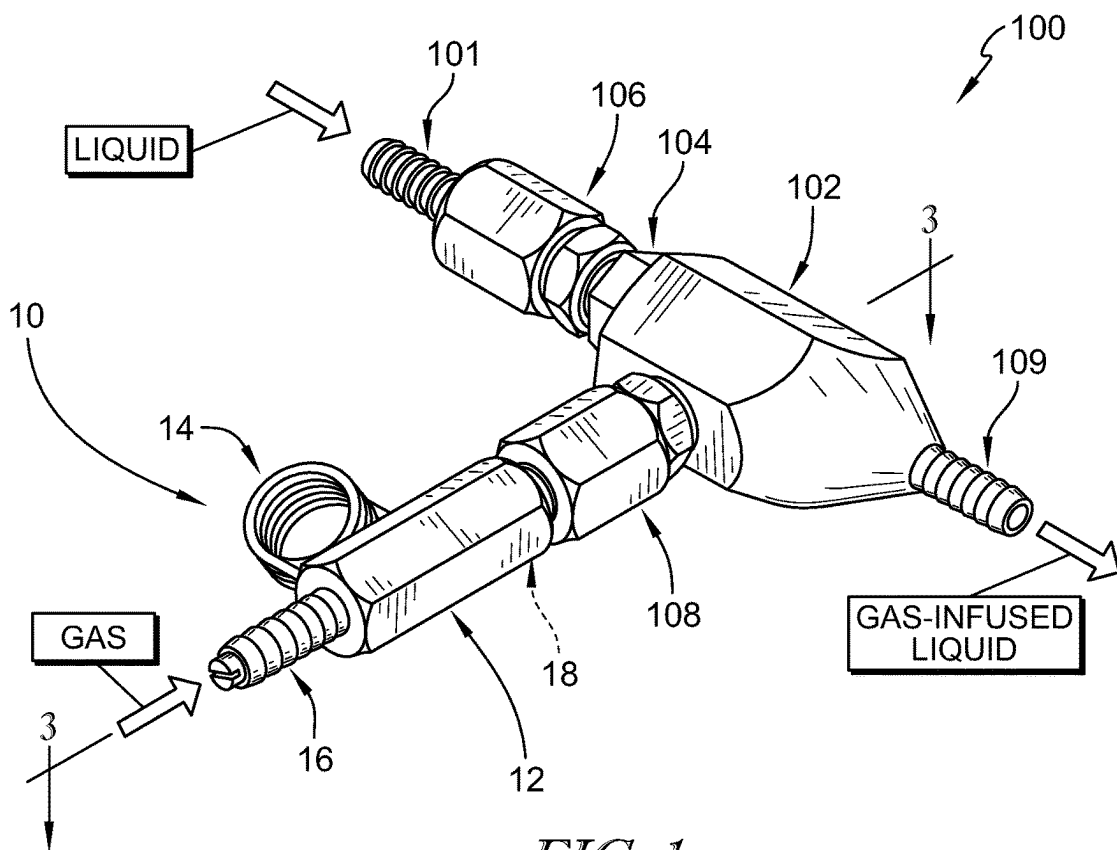


FIG. 1

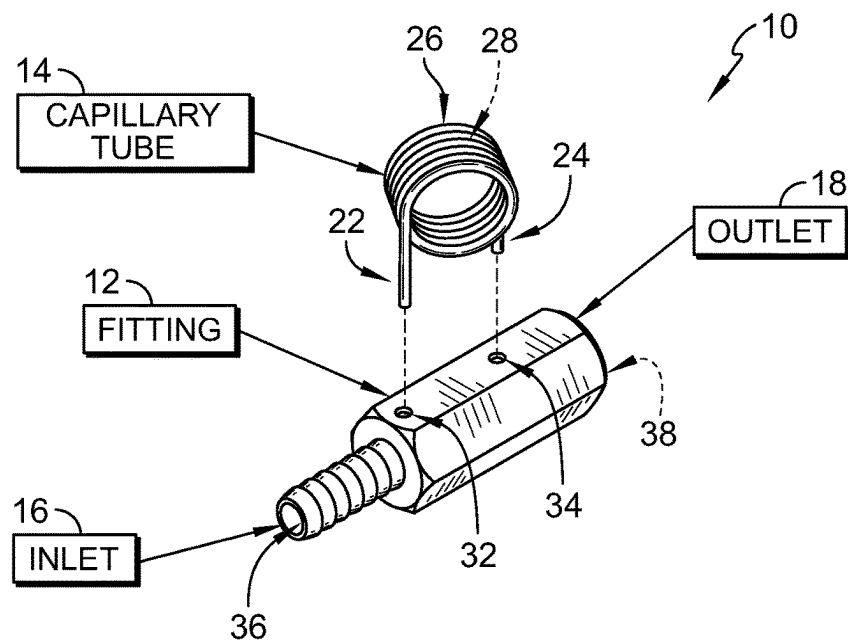
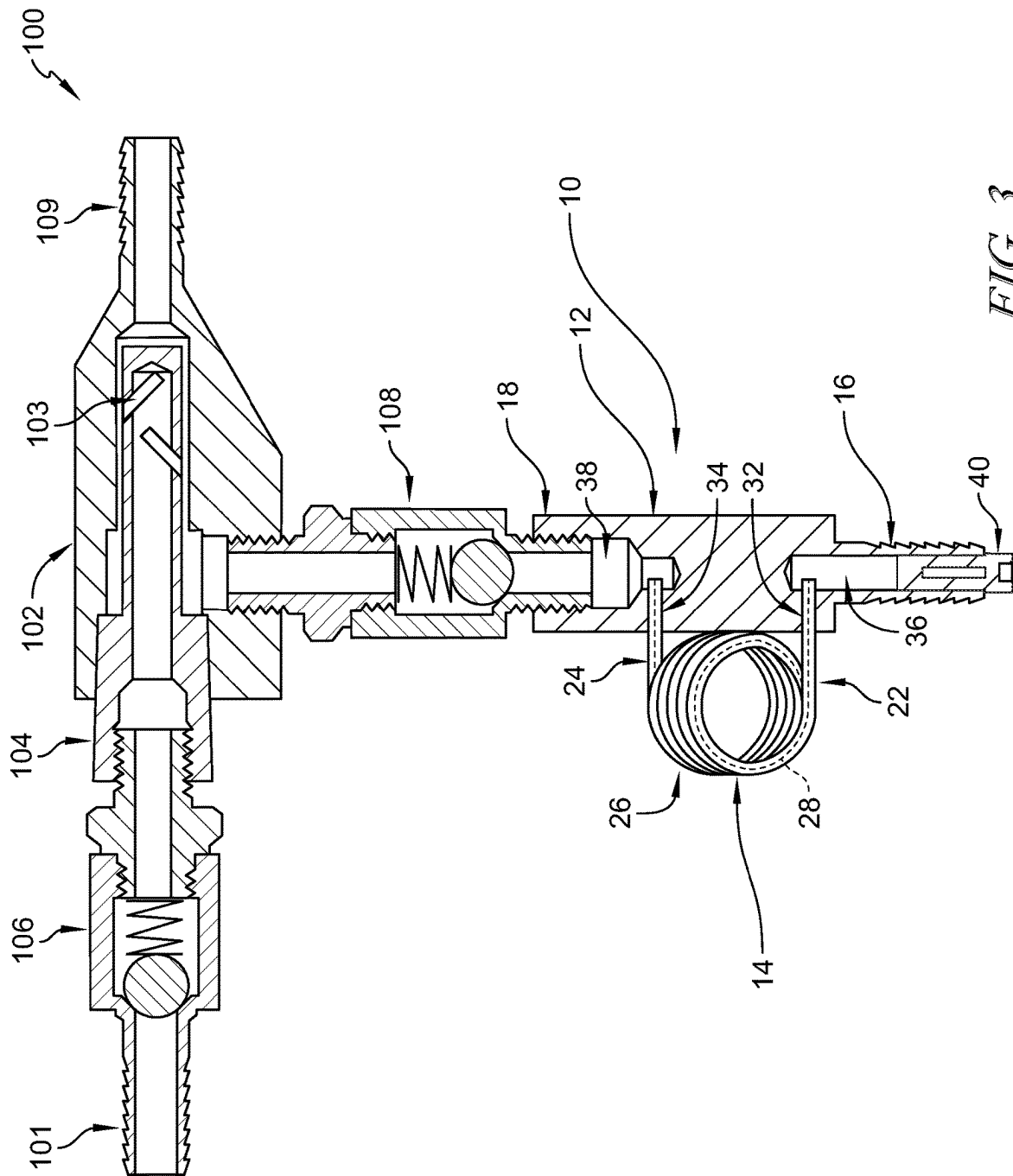


FIG. 2



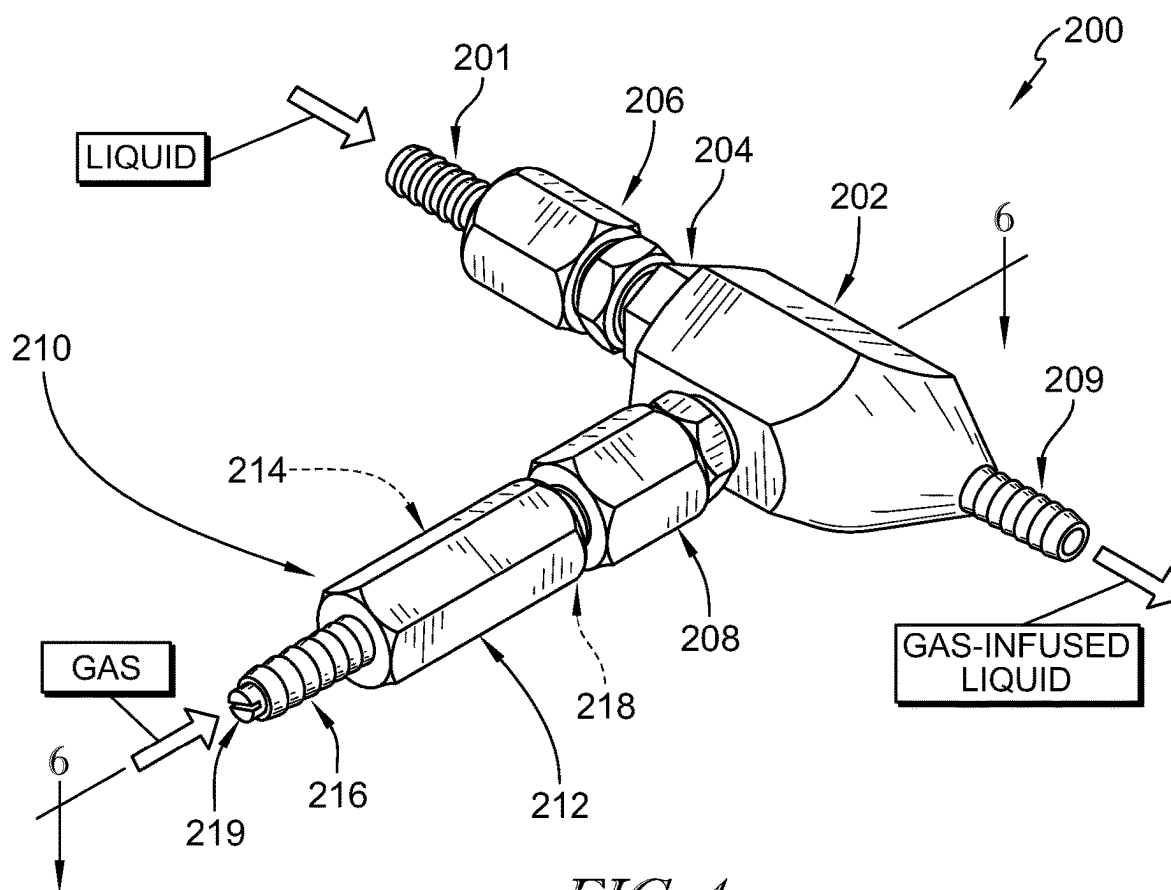


FIG. 4

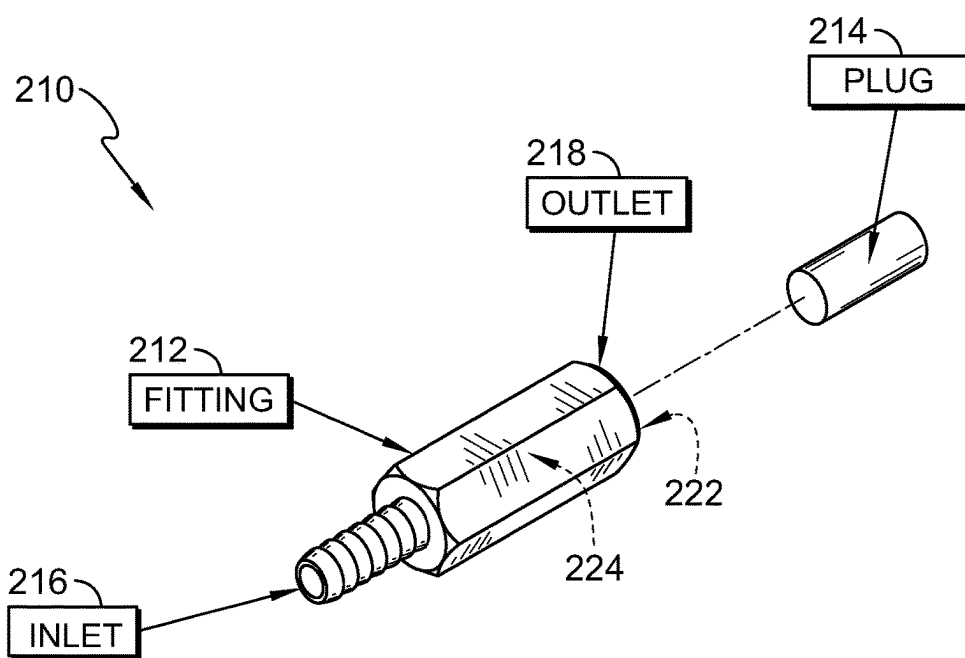
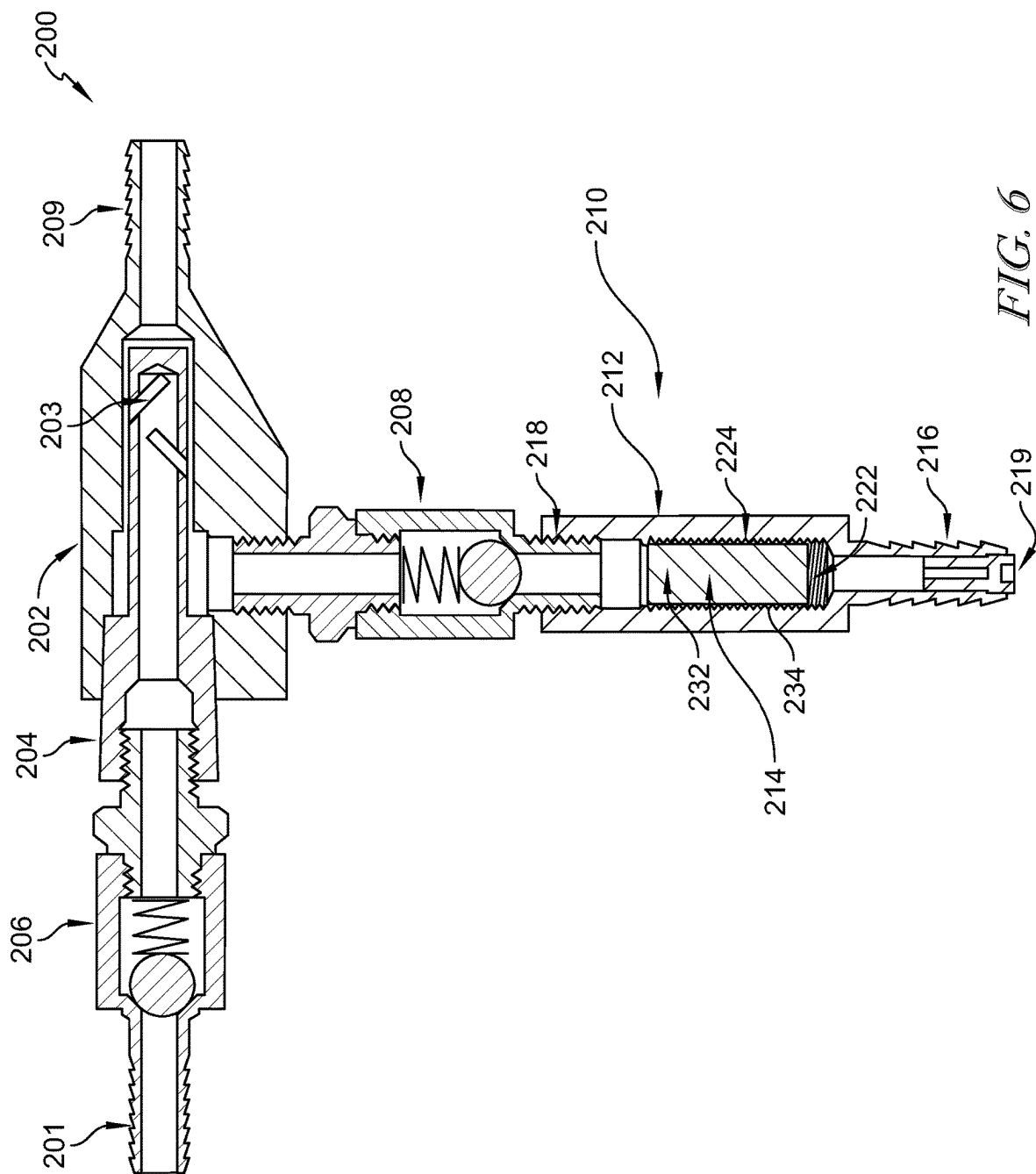


FIG. 5



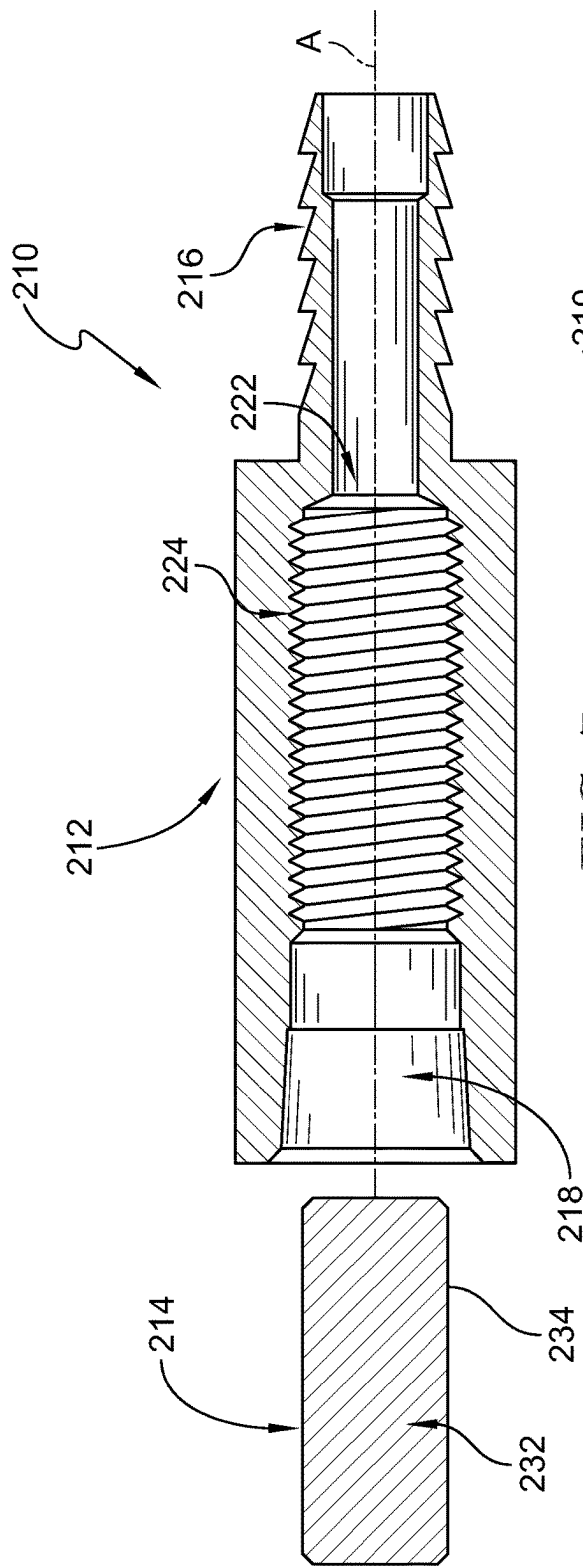


FIG. 7

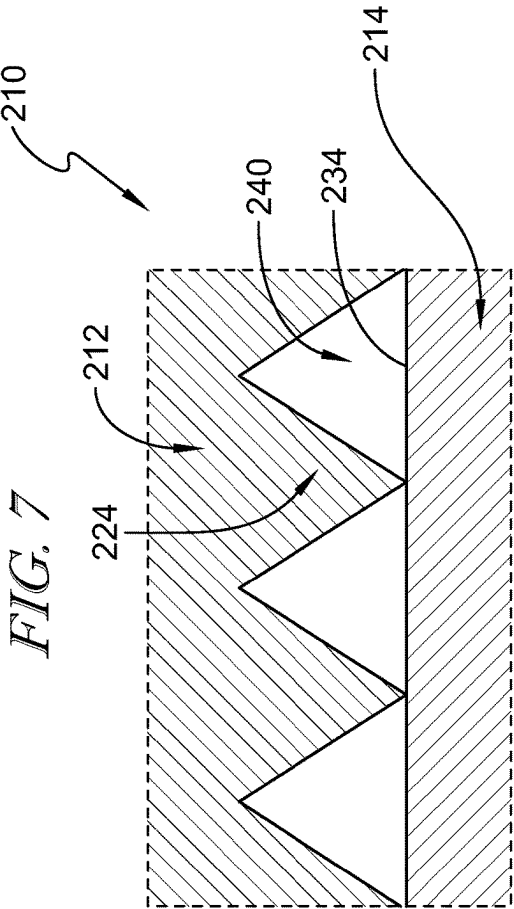


FIG. 8

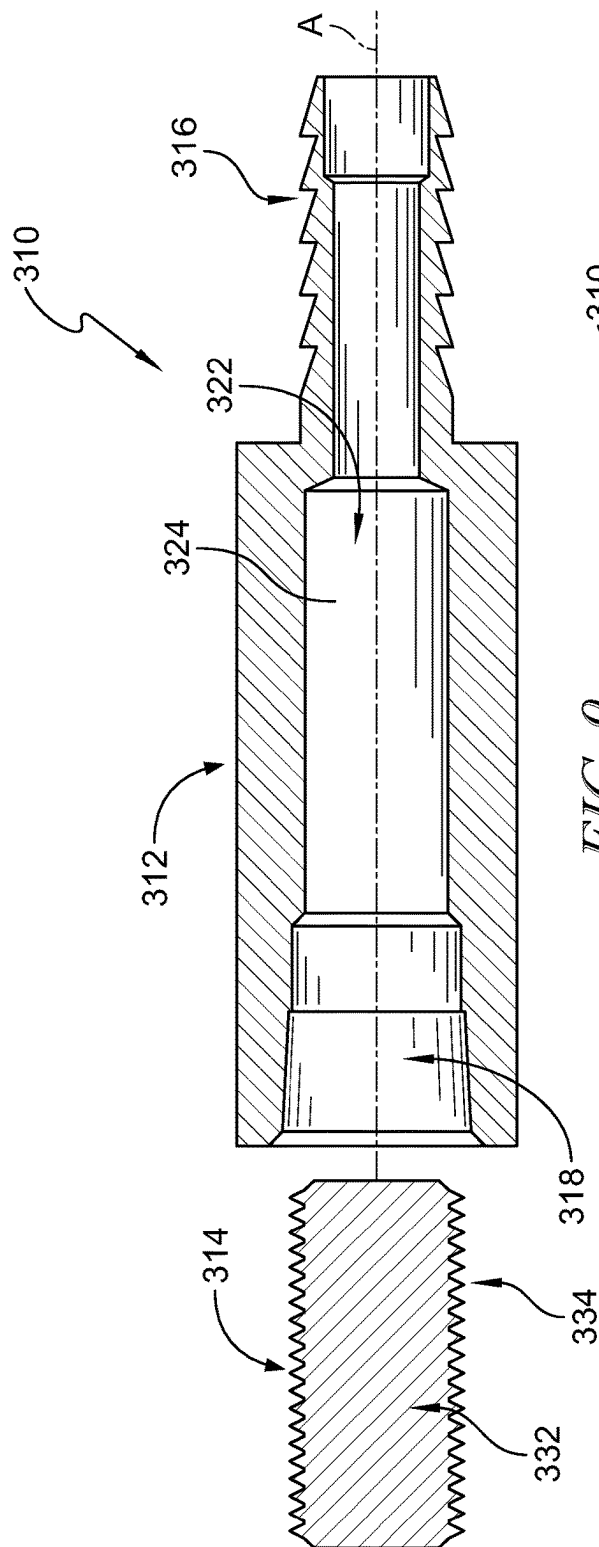


FIG. 9

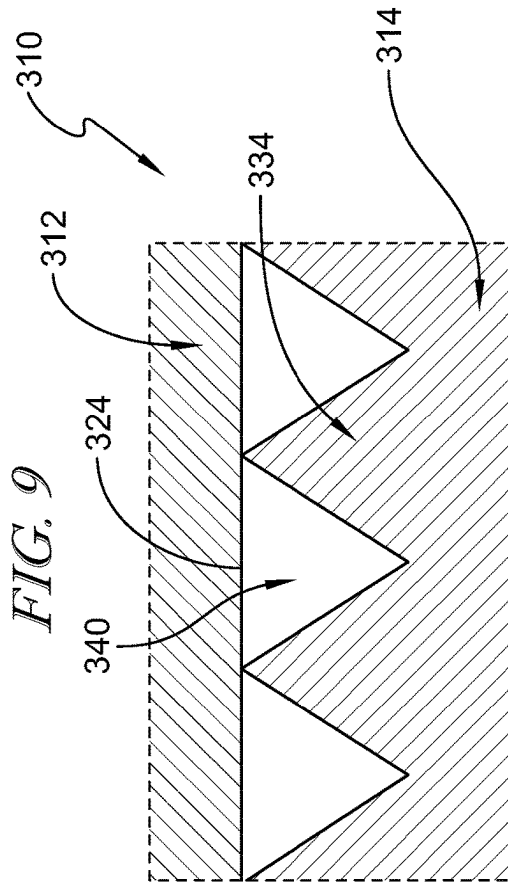
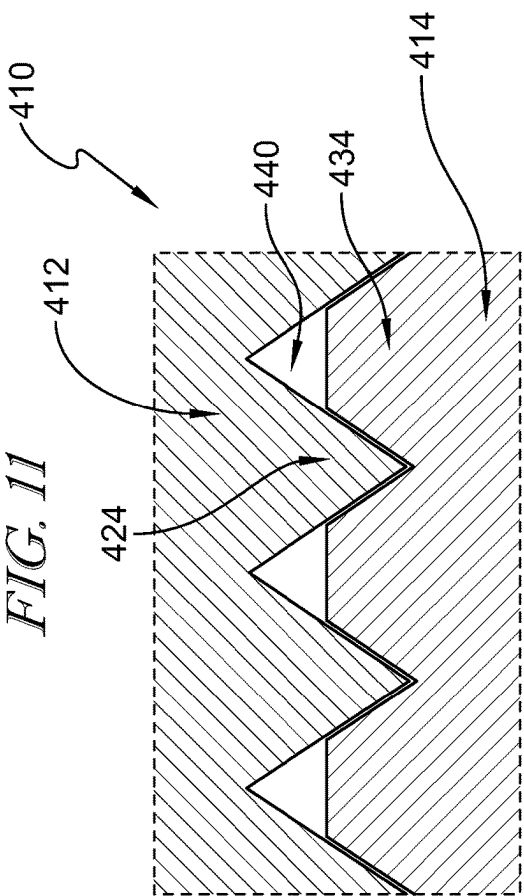
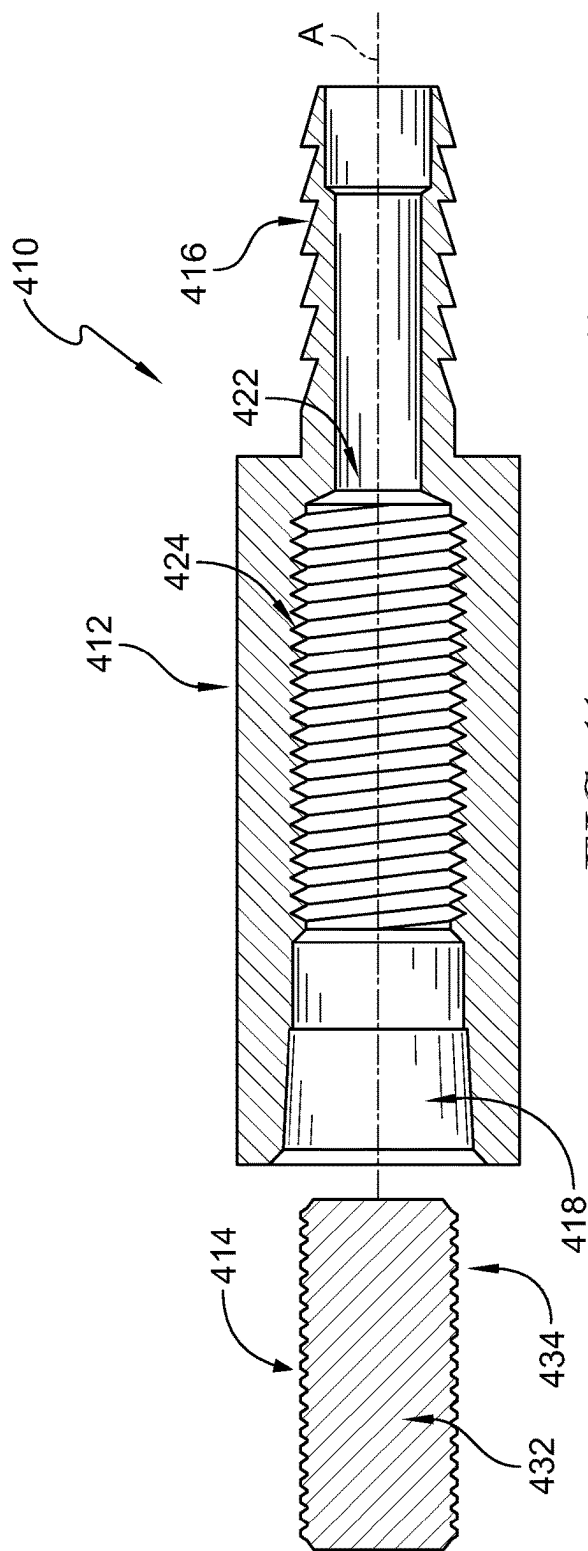


FIG. 10



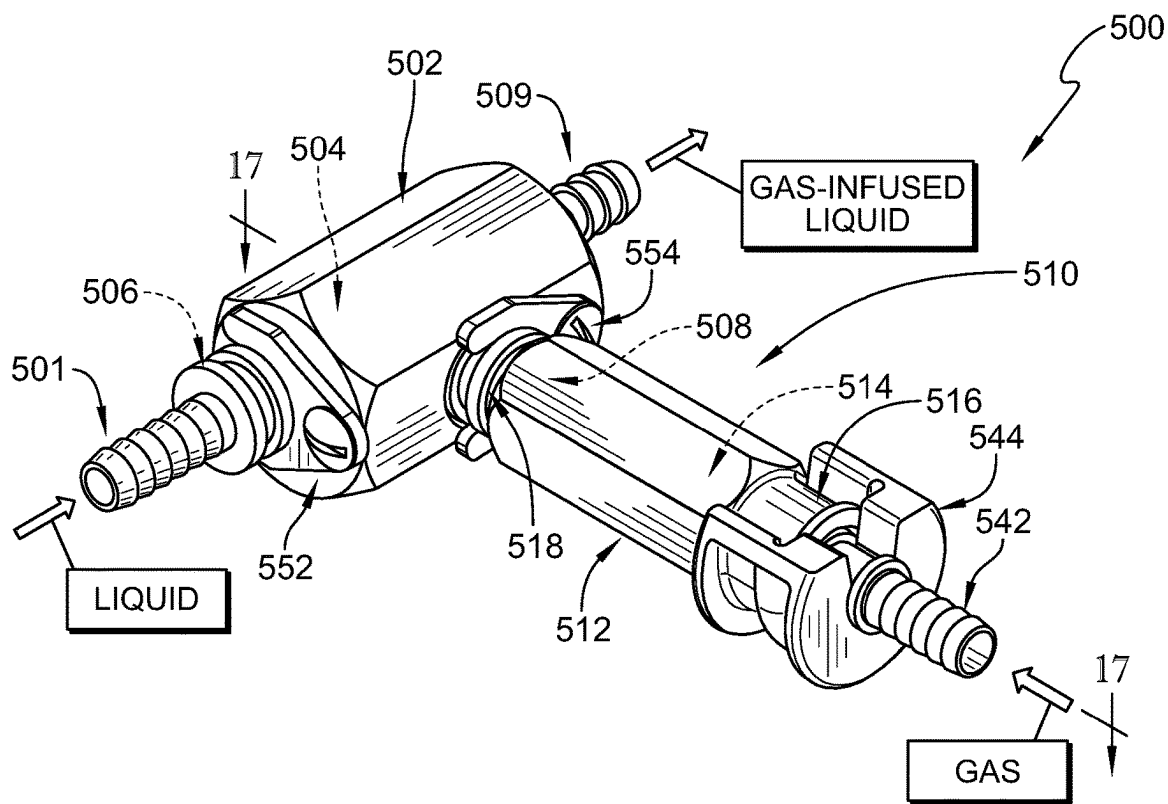


FIG. 13

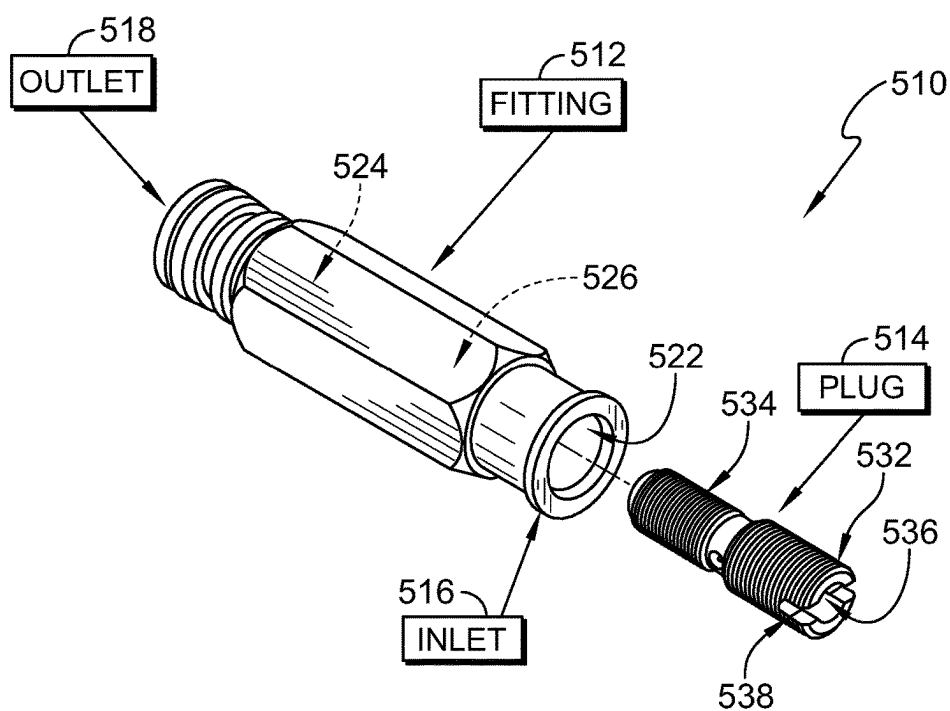


FIG. 14

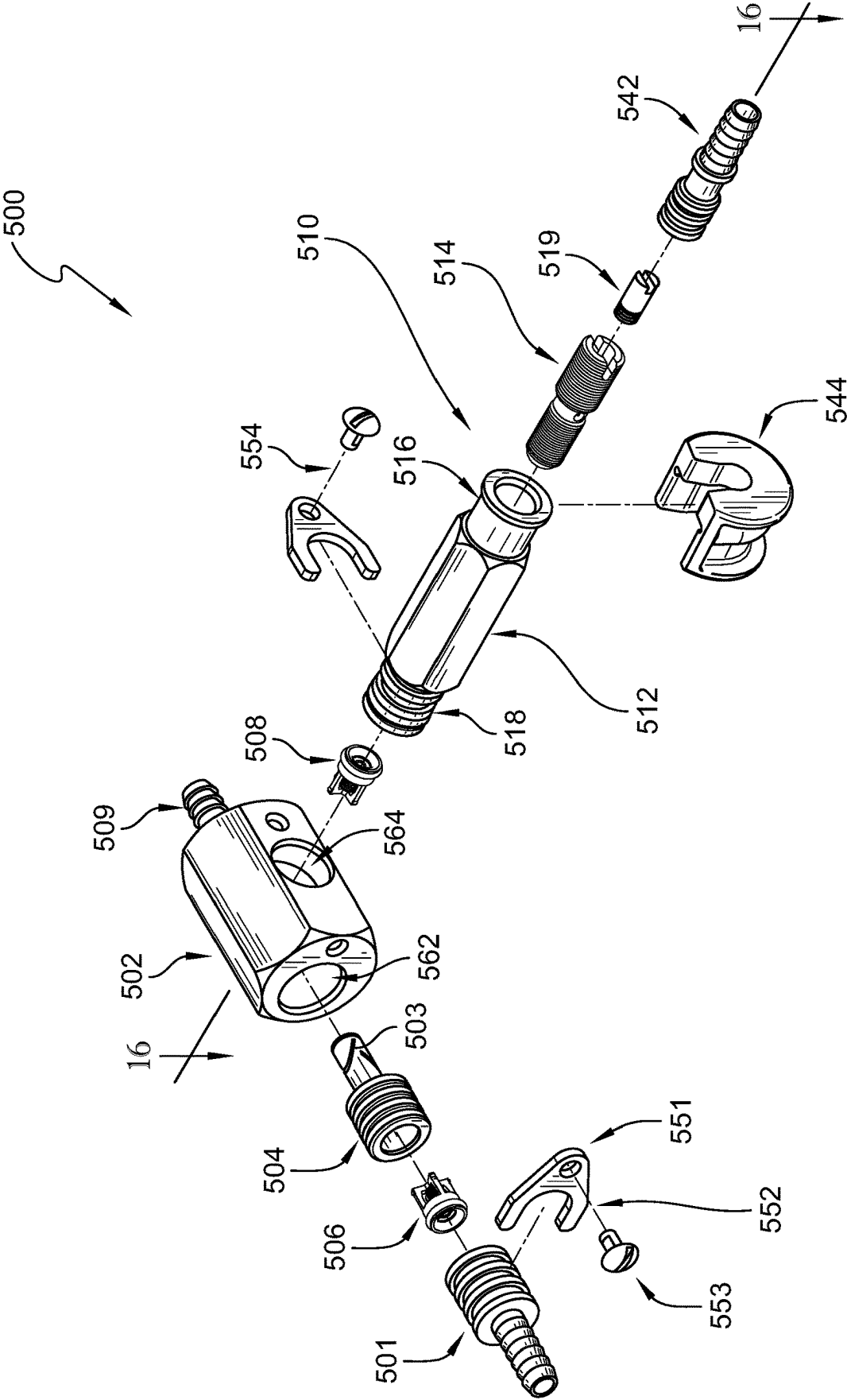


FIG. 15

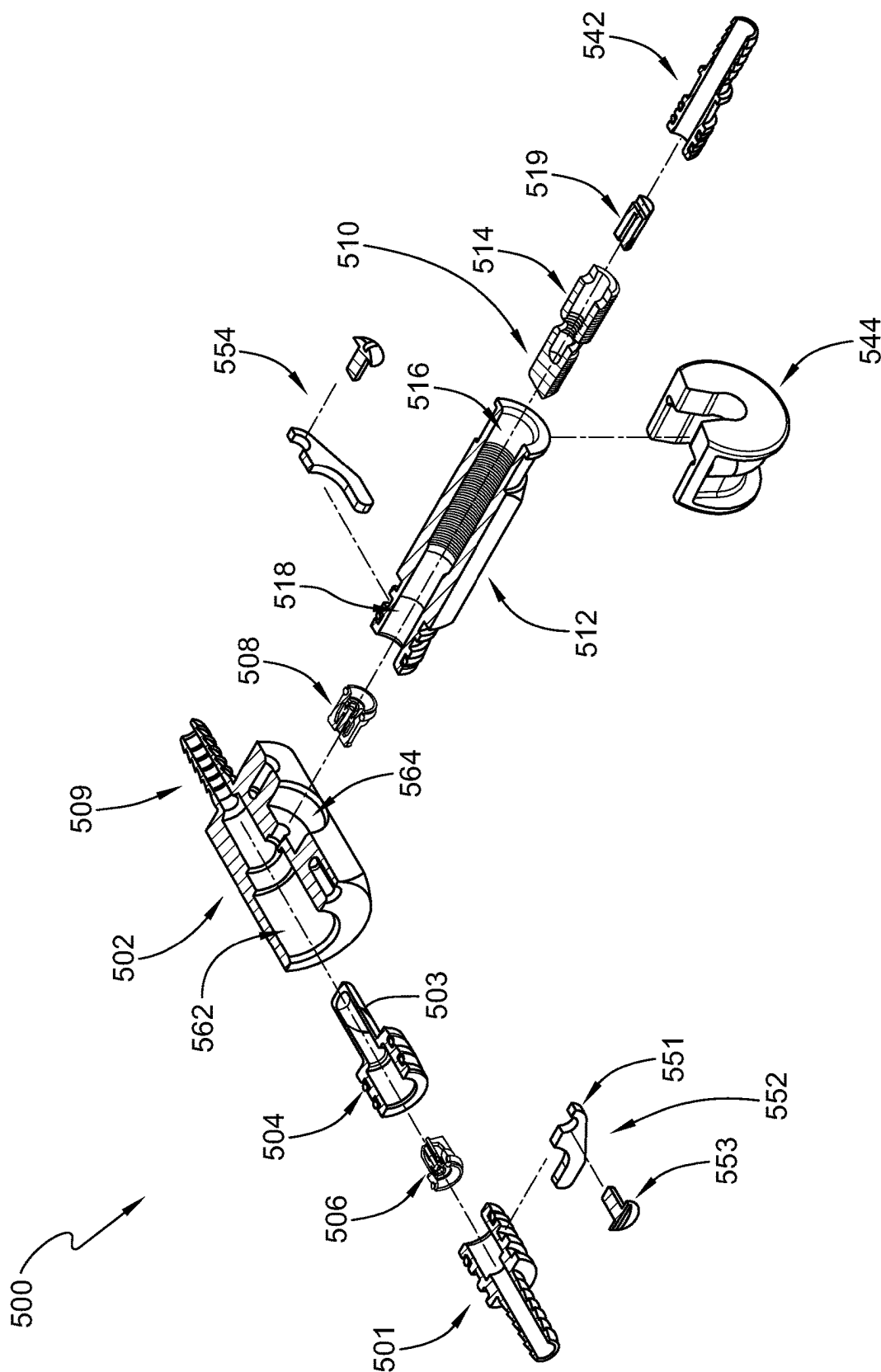


FIG. 16

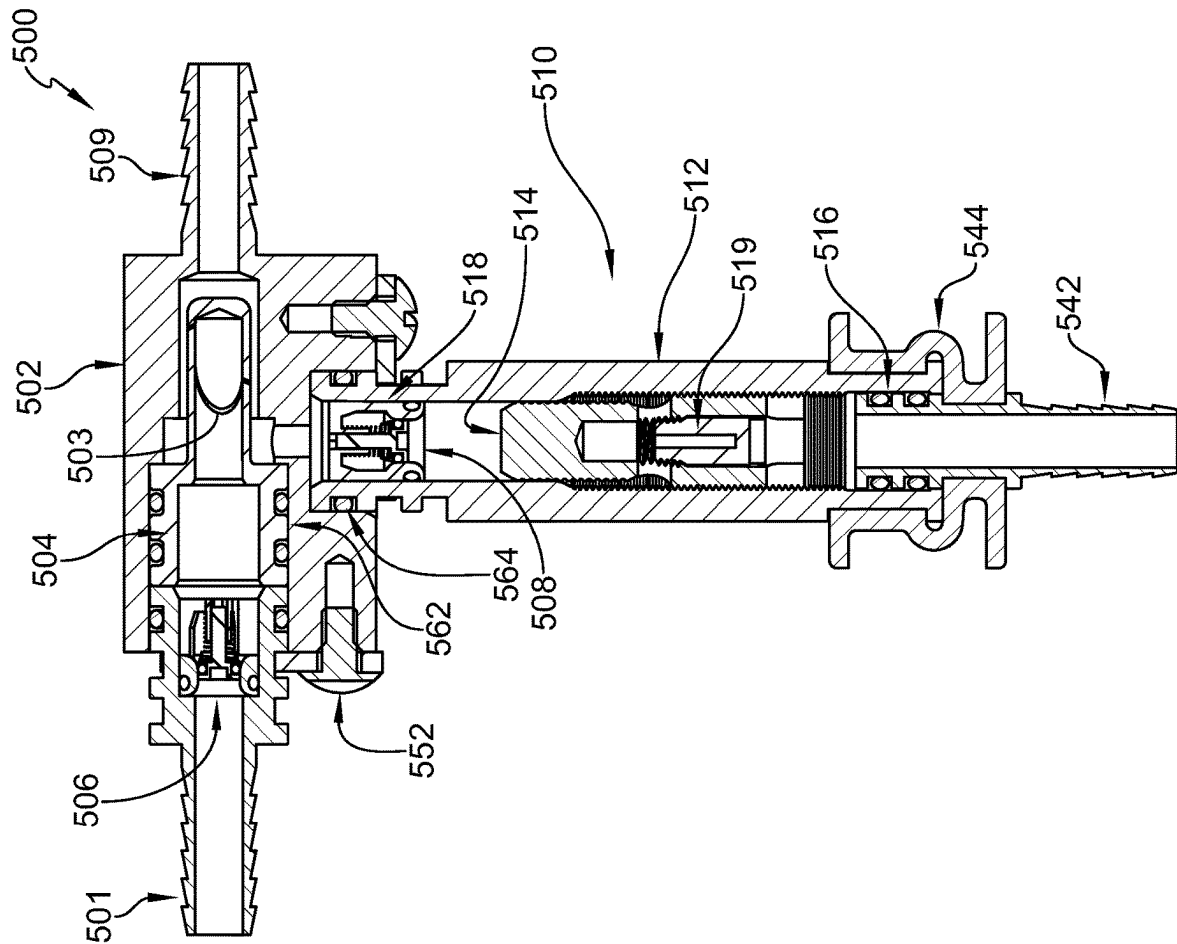


FIG. 17

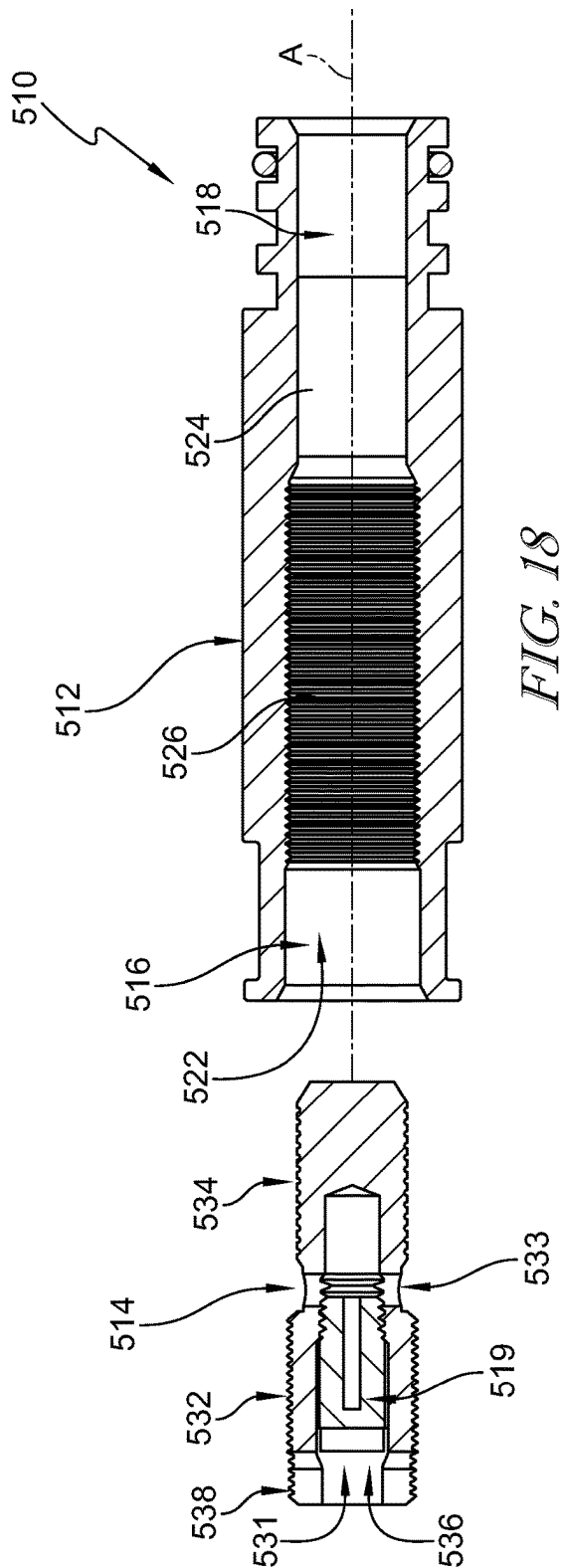


FIG. 18

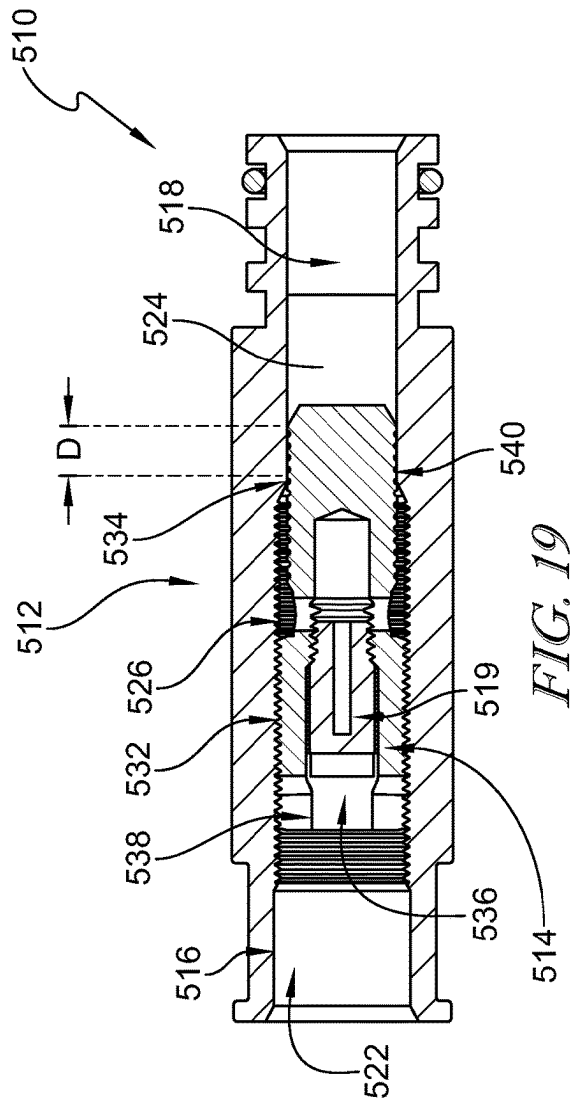
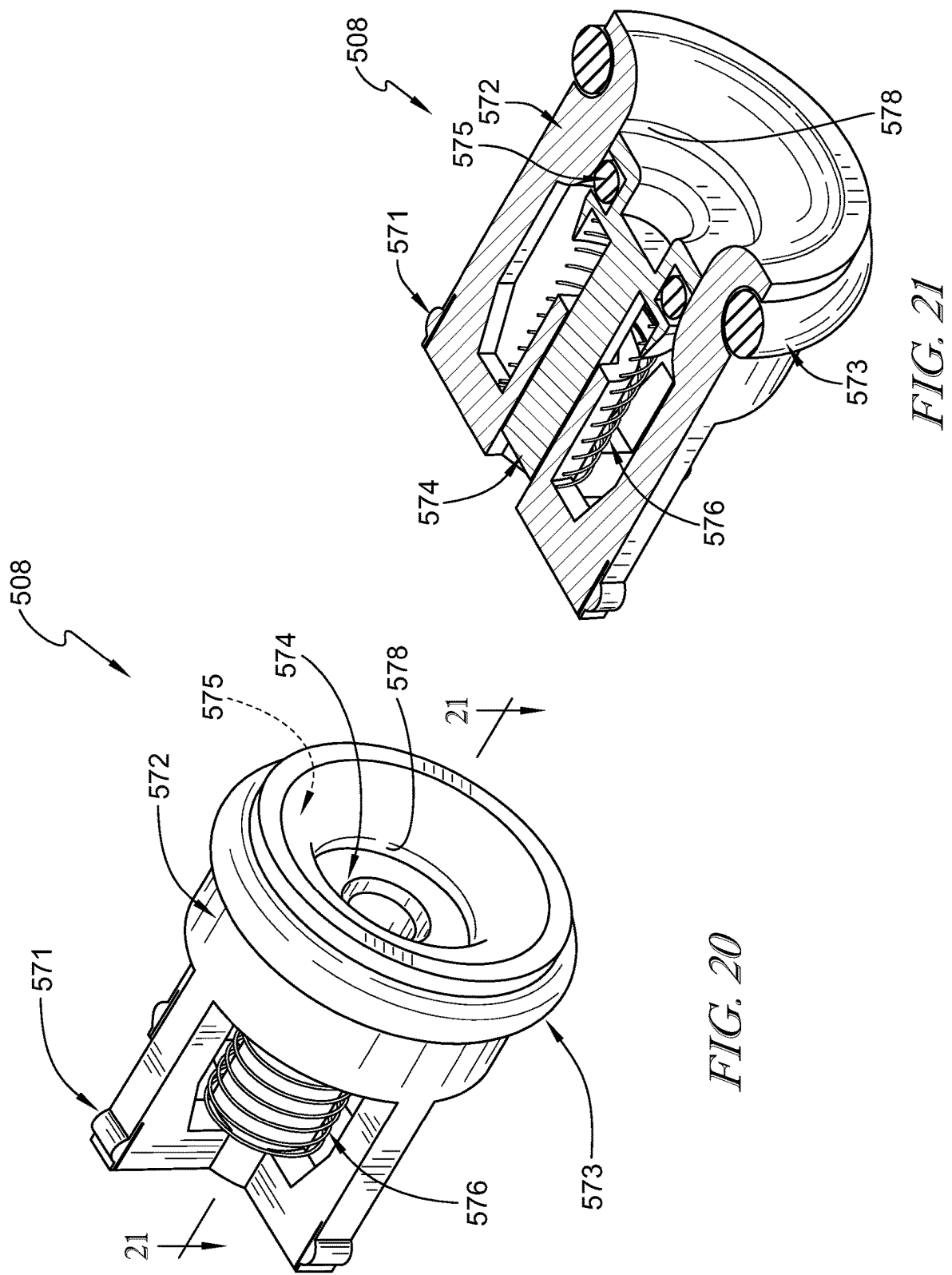


FIG. 19



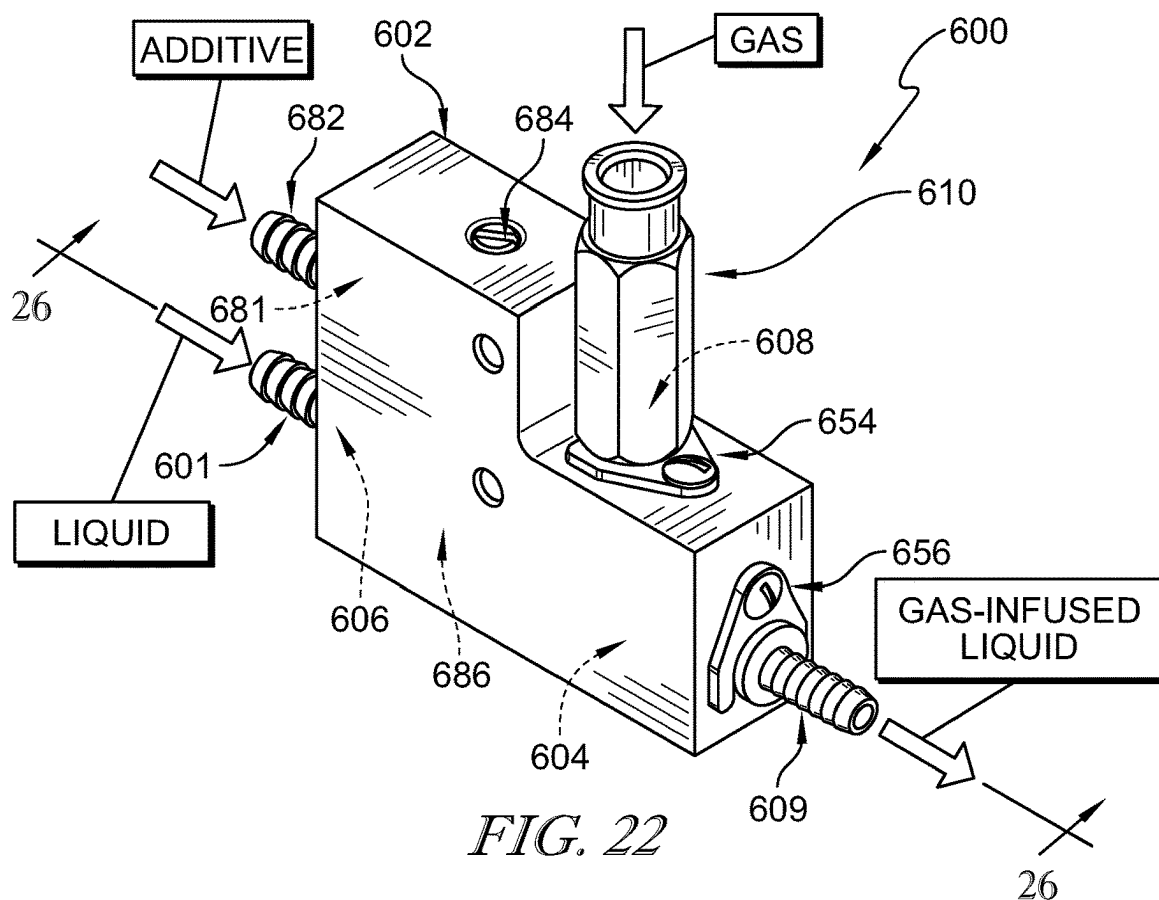


FIG. 22

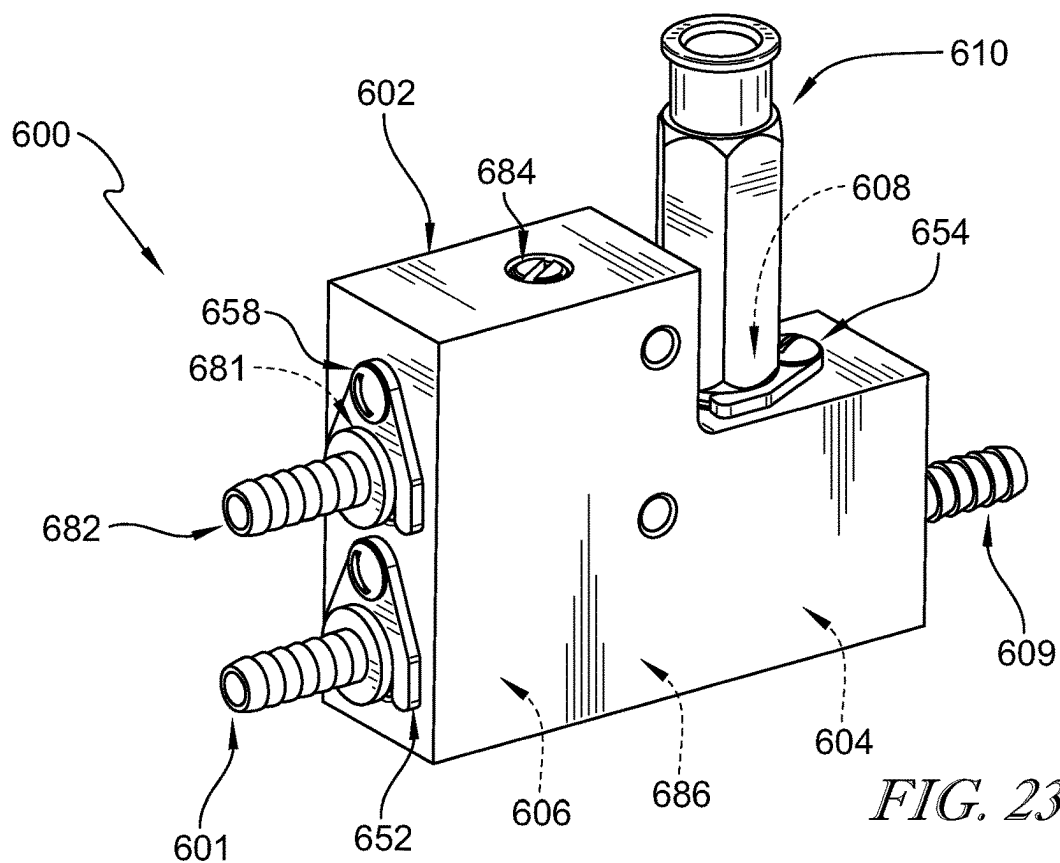


FIG. 23

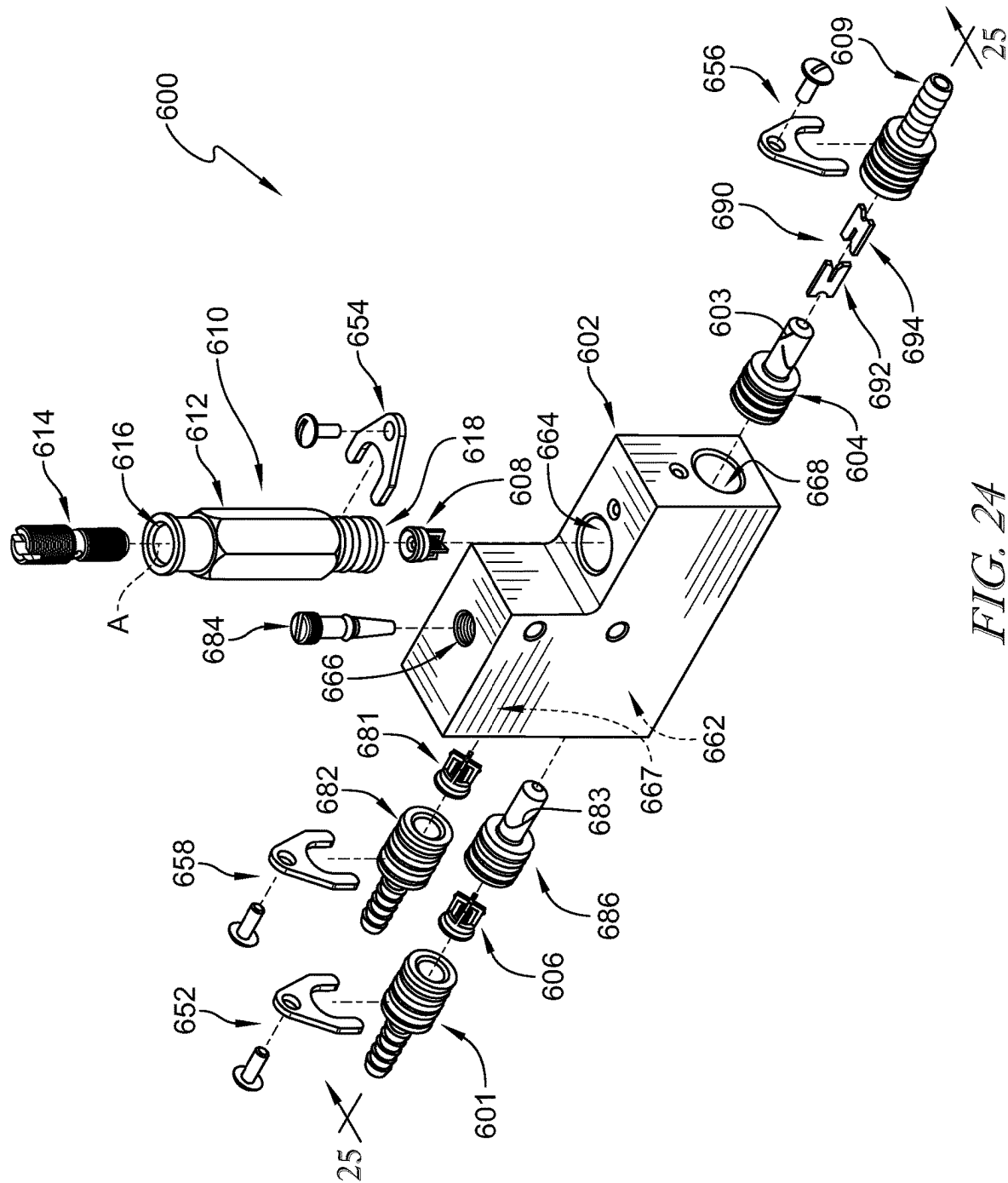


FIG. 24

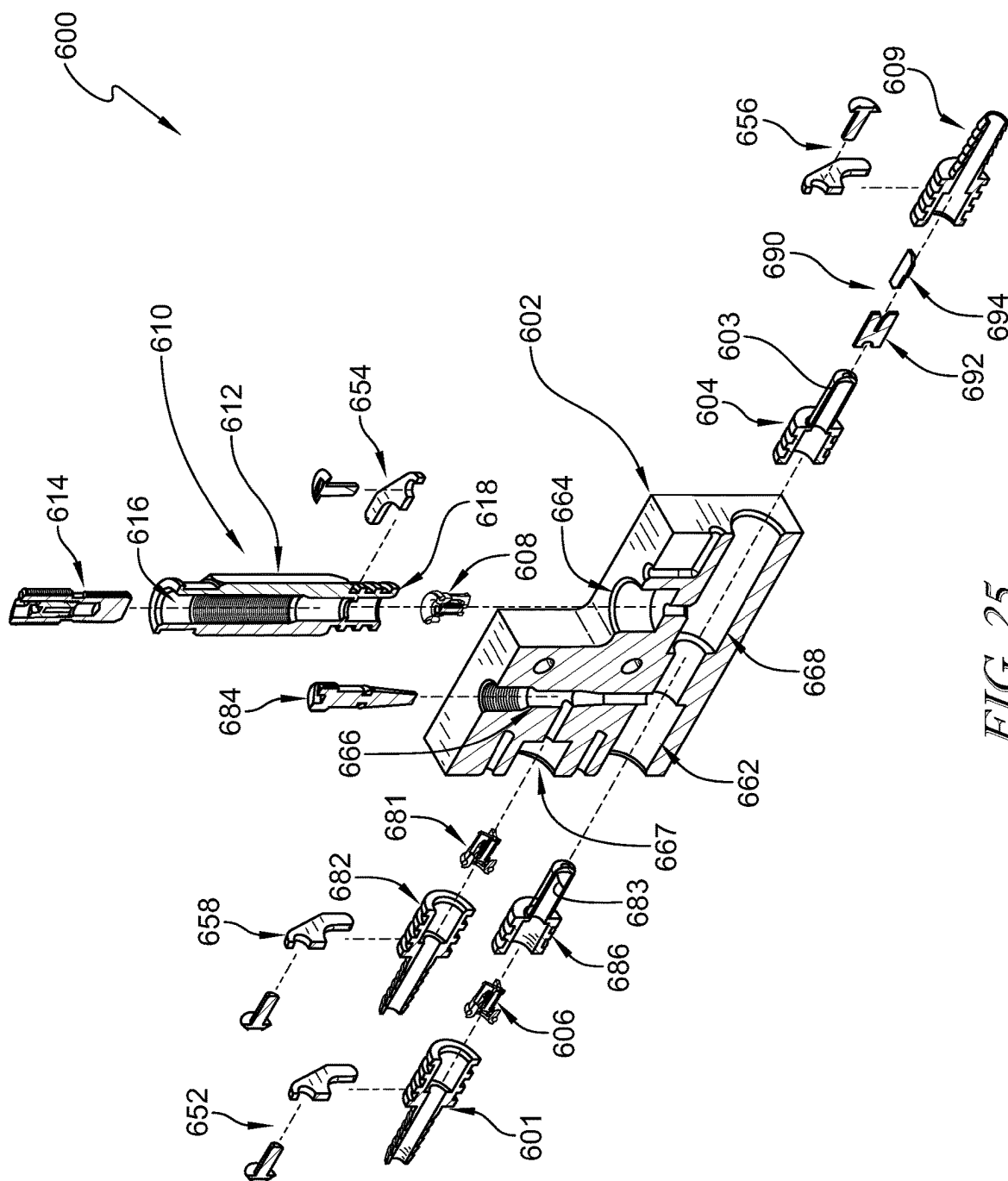


FIG. 25

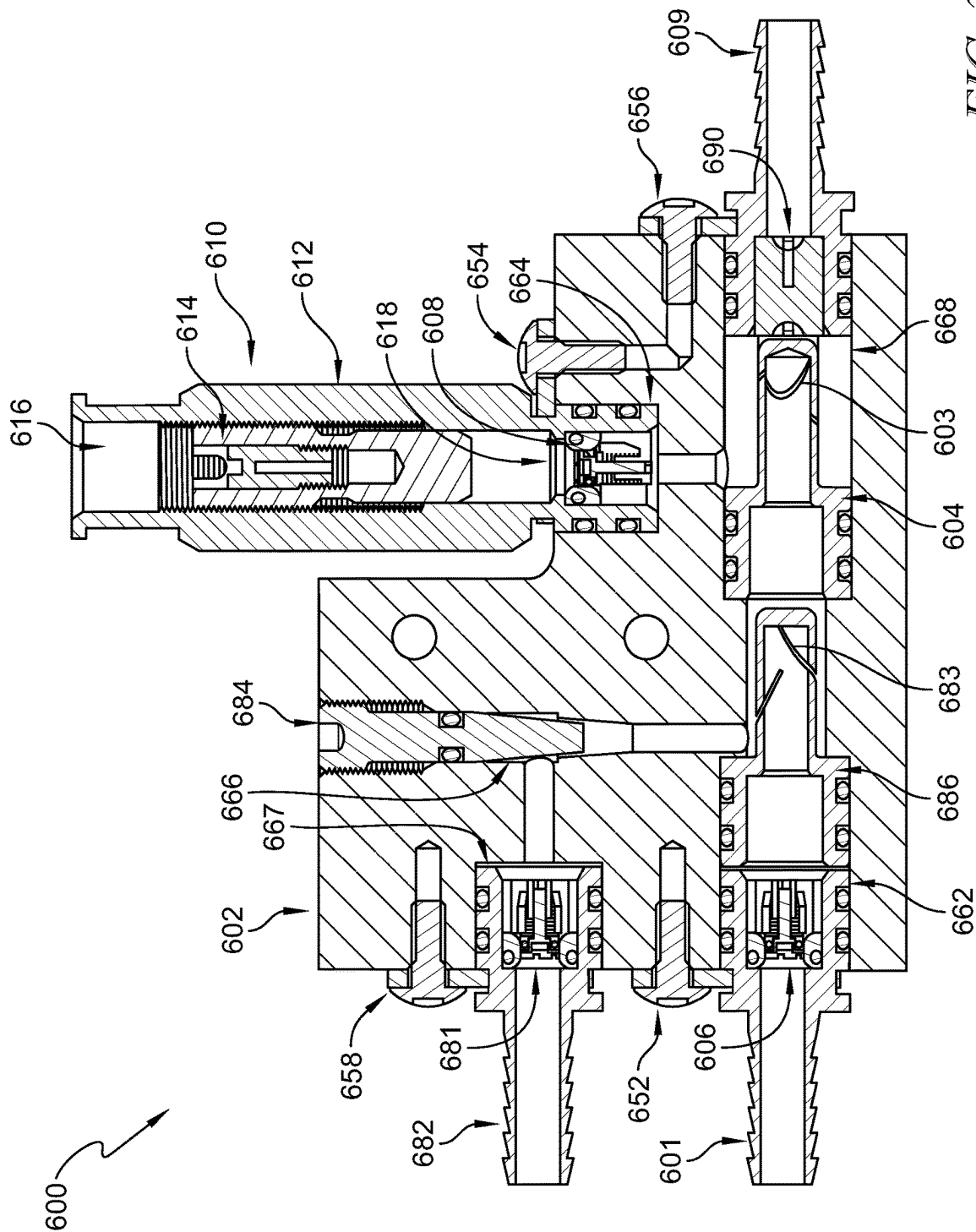


FIG. 26

1

GAS INFUSER FOR LIQUIDS**CROSS REFERENCE TO RELATED APPLICATIONS**

This application is a Continuation of copending U.S. patent application Ser. No. 16/968,238, filed Aug. 7, 2020, which is a U.S. nationalization under 35 U.S.C. § 371 of International Application No. PCT/US2019/016781, filed Feb. 6, 2019, which claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Application Ser. Nos. 62/628,035, filed Feb. 8, 2018, and 62/644,026, filed Mar. 16, 2018, the disclosures of which are expressly incorporated by reference herein in their entirety.

BACKGROUND

The present disclosure relates to gas infusers, and particularly to gas infusers for forming a gas-infused liquid. More particularly, the present disclosure relates to gas infusers for forming a gas-infused liquid used in beverages.

SUMMARY

According to the present disclosure, a gas infuser includes a mixing chamber and a liquid injector. Gas is supplied into the mixing chamber and liquid flows through openings of the injector to mix with the gas in the mixing chamber and form a gas-infused liquid.

In illustrative embodiments, the gas is supplied through a gas-volume controller coupled to the mixing chamber. The gas-volume controller includes a fitting and a capillary tube. An inlet of the fitting is separated from an outlet of the fitting such that gas passes from the inlet through the capillary tube to the outlet. A circuitous path is formed through the capillary tube to control a volume of gas that passes from the inlet to the outlet of the fitting substantially independent of pressure fluctuations of the gas at the inlet.

In illustrative embodiments, the capillary tube includes a coil and stems extending from opposing ends of the coil. One stem is positioned in an opening associated with the inlet of the fitting and the other stem is positioned in an opening associated with the outlet. The outlet is coupled to the mixing chamber.

According to another aspect of the present disclosure, the gas is supplied through a gas-volume controller coupled to the mixing chamber. The gas-volume controller includes a fitting and a plug. The plug is inserted into the fitting to separate an inlet of the fitting from an outlet. Internal threads of the fitting engage with the plug to form a circuitous path from the inlet to the outlet to control a volume of gas that passes through the gas-volume controller substantially independent of pressure fluctuations of the gas at the inlet.

In illustrative embodiments, an exterior of the plug is threaded and an interior of the fitting is smooth. The threads of the plug engage with the interior of the fitting to form the circuitous path.

In illustrative embodiments, an exterior of the plug is threaded and an interior of the fitting is threaded. The threads of the plug and fitting engage with one another. Peaks of the threads on the plug are removed such that the threads of the plug extend partially into the threads of the fitting. The circuitous path is formed between the threads of the plug and fitting.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure will be described hereafter with reference to the attached drawings which are given as a non-limiting example only, in which:

2

FIG. 1 is a perspective view of a gas infuser in accordance with the present disclosure showing that the gas infuser includes a mixing chamber, a liquid injector, and a gas-volume controller and suggesting that liquid is supplied to the mixing chamber through the injector and gas is supplied to the mixing chamber through the gas-volume controller for mixing with the liquid and forming a gas-infused liquid;

FIG. 2 is an exploded assembly view of the gas-volume controller of FIG. 1 showing that the gas-volume controller includes a fitting having an inlet and an outlet and a capillary tube coupled to the fitting and suggesting that the capillary tube controls a volume of gas that passes from the inlet to the outlet of the fitting substantially independent of pressure fluctuations of the gas at the inlet;

FIG. 3 is a sectional view taken along line 3-3 in FIG. 1 showing that the capillary tube includes a coil and stems extending from opposing ends of the coil with one stem positioned in an opening associated with the inlet of the fitting and the other stem positioned in an opening associated with the outlet and suggesting that a circuitous path is formed through the stems and coil to direct gas from the inlet to the outlet;

FIG. 4 is a perspective view of another embodiment of a gas infuser in accordance with the present disclosure showing that the gas infuser includes a mixing chamber, a liquid injector, and a gas-volume controller and suggesting that liquid is supplied to the mixing chamber through the injector and gas is supplied to the mixing chamber through the gas-volume controller for mixing with the liquid and forming a gas-infused liquid;

FIG. 5 is an exploded assembly view of the gas-volume controller of FIG. 4 showing that the gas-volume controller includes a fitting having an inlet and an outlet and a plug coupled to the fitting and suggesting that the plug engages with threads of an internal cavity of the fitting, as shown in FIG. 6, to form a circuitous path from the inlet to the outlet that controls a volume of gas that passes through the gas-volume controller substantially independent of pressure fluctuations of the gas at the inlet;

FIG. 6 is a sectional view taken along line 6-6 in FIG. 4 showing the plug received in the fitting such that the inlet is separated from the outlet and suggesting that gas enters the inlet and flows along the threads around the plug to the outlet;

FIG. 7 is a sectional view of the gas-volume controller of FIG. 6 showing the plug aligned with the fitting and suggesting that the plug is pressed into the fitting to engage with the threads to define the circuitous path between the fitting and the plug;

FIG. 8 is an enlarged view of the gas-volume controller of FIG. 6 showing the threads of the fitting engaged with the exterior surface of the plug to form a seal such that adjacent portions of the circuitous path are separated from one another to force the gas to travel around the plug along the circuitous path;

FIG. 9 is a sectional view of another embodiment of a gas-volume controller in accordance with the present disclosure showing that the gas-volume controller includes a fitting and a plug and suggesting that the plug is pressed into the fitting to engage threads of the plug with an interior surface of the fitting to define a circuitous path between an inlet of the fitting to an outlet;

FIG. 10 is an enlarged view of the gas-volume controller of FIG. 9 showing the plug received in the fitting and the threads of the plug engaged with the interior surface of the fitting to form a seal such that adjacent portions of the

3

circuitous path are separated from one another to force gas to travel around the plug along the circuitous path;

FIG. 11 is a sectional view of another embodiment of a gas-volume controller in accordance with the present disclosure showing that the gas-volume controller includes a fitting and a plug and suggesting that the plug is received in the fitting to engage threads of the plug with threads of the fitting to define a circuitous path between an inlet of the fitting to an outlet;

FIG. 12 is an enlarged view of the gas-volume controller of FIG. 11 showing the plug received in the fitting and the threads of the plug engaged with the threads of the fitting to form a seal such that adjacent portions of the circuitous path are separated from one another to force gas to travel around the plug along the circuitous path

FIG. 13 is a perspective view of another embodiment of a gas infuser in accordance with the present disclosure showing that the gas infuser includes a mixing chamber, a liquid injector, and a gas-volume controller and suggesting that liquid is supplied to the mixing chamber through the injector and gas is supplied to the mixing chamber through the gas-volume controller for mixing with the liquid and forming a gas-infused liquid;

FIG. 14 is an exploded assembly view of the gas-volume controller of FIG. 13 showing that the gas-volume controller includes a fitting having an inlet and an outlet and a plug coupled to the fitting and suggesting that a threaded head of the plug engages with an internal cavity of the fitting, as shown in FIG. 15, to form a circuitous path from the inlet to the outlet that controls a volume of gas that passes through the gas-volume controller substantially independent of pressure fluctuations of the gas at the inlet;

FIG. 15 is an exploded assembly view of the gas infuser of FIG. 13;

FIG. 16 is a sectional view taken along line 16-16 in FIG. 15;

FIG. 17 is a sectional view taken along line 17-17 in FIG. 13 showing the plug received in the fitting such that the inlet is separated from the outlet and suggesting that gas enters the inlet and flows along the threads around the plug to the outlet;

FIG. 18 is a sectional view of the gas-volume controller of FIG. 17 showing the plug aligned with the fitting and suggesting that the plug is received in the fitting to engage the threads of the head of the plug with an inner surface of the cavity to define the circuitous path between the fitting and the plug;

FIG. 19 is a view similar to FIG. 18 showing the plug received in the fitting and suggesting that a depth (D) of the plug relative to the fitting defines a length of the circuitous path and that the depth is adjustable at the selection of a user;

FIG. 20 is a perspective view of a check valve in accordance with the present disclosure showing that the check valve includes a frame and a plunger movable relative to an opening in the frame;

FIG. 21 is a sectional view taken along line 21-21 in FIG. 20;

FIG. 22 is a front perspective view of another embodiment of a gas infuser in accordance with the present disclosure showing that the gas infuser includes a mixing chamber, liquid injectors, and a gas-volume controller and suggesting that liquid is supplied to the mixing chamber through a first injector for mixing of the liquid with an additive and gas is supplied to the mixing chamber through the gas-volume controller for mixing with the liquid-additive solution provided through a second injector to form a gas-infused liquid;

4

FIG. 23 is a rear perspective view of the gas infuser of FIG. 22;

FIG. 24 is an exploded assembly view of the gas infuser of FIG. 22;

FIG. 25 is a sectional view taken along line 25-25 in FIG. 24 showing that the mixing chamber includes a plurality of intersecting cavities for directing the flow of liquid, additive, and gas toward an outlet of the gas infuser; and

FIG. 26 is a sectional view taken along line 26-26 in FIG. 22 showing the injectors received in the cavities of the mixing chamber and suggesting that a flow adjuster controls an amount of additive supplied for mixing with the liquid at the first injector.

The exemplification set out herein illustrates embodiments of the disclosure that are not to be construed as limiting the scope of the disclosure in any manner. Additional features of the present disclosure will become apparent to those skilled in the art upon consideration of the following detailed description of illustrative embodiments exemplifying the best mode of carrying out the disclosure as presently perceived.

DETAILED DESCRIPTION

While the present disclosure may be susceptible to embodiment in different forms, there is shown in the drawings, and herein will be described in detail, embodiments with the understanding that the present description is to be considered an exemplification of the principles of the disclosure. The disclosure is not limited in its application to the details of structure, function, construction, or the arrangement of components set forth in the following description or illustrated in the drawings. The disclosure is capable of other embodiments and of being practiced or of being carried out in various ways. In addition, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of various phrases and terms is meant to encompass the items or functions identified and equivalents thereof as well as additional items or functions. Unless limited otherwise, various phrases, terms, and variations thereof herein are used broadly and encompass all variations of such phrases and terms. Furthermore, and as described in subsequent paragraphs, the specific configurations illustrated in the drawings are intended to exemplify embodiments of the disclosure. However, other alternative structures, functions, and configurations are possible which are considered to be within the teachings of the present disclosure. Furthermore, unless otherwise indicated, the term "or" is to be considered inclusive.

The term beverage as may be used herein is intended to be broadly defined as including, but not limited to, brewed beverages such as coffee and tea, juices, and diluted concentrates of brewed beverages, juices, or other syrups, as well as any other beverages or consumable liquids. The term gas as may be used herein is intended to be broadly defined as including, but not limited to, oxygen, air, carbon dioxide (CO₂), nitrogen (N₂), as well as any other gaseous substance. The foregoing terms as well as other terms should be broadly interpreted throughout this application to include all known as well as all hereafter discovered versions, equivalents, variations and other forms of the abovementioned terms as well as other terms. The present disclosure is intended to be broadly interpreted and not limited.

A gas infuser 100 in accordance with the present disclosure is shown in FIG. 1. Gas infuser 100 includes a mixing chamber 102 and a liquid injector 104 coupled to mixing

5

chamber 102. A check valve 106 is coupled to injector 104 and provides an inlet 101 for passing a liquid through injector 104 into mixing chamber 102. In the illustrative embodiment, a gas-volume controller 10 in accordance with the present disclosure is coupled to mixing chamber 102 to control a volume of gas delivered to mixing chamber 102. Gas-volume controller 10 includes a fitting 12 and a capillary tube 14 coupled to fitting 12 as shown in FIGS. 1 and 2. Fitting 12 has an inlet 16 and an outlet 18. Outlet 18 is coupled to check valve 108.

Injector 104 is received in mixing chamber 102 and includes a plurality of slots 103 to allow liquid to pass through injector 104 into mixing chamber 102 as suggested in FIG. 3. A liquid, such as water or pre-formed beverage for example, passes from inlet 101 through check valve 106 and injector 104 into mixing chamber 102 as suggested in FIG. 1. A gas, such as CO₂ or N₂, passes from inlet 16 through capillary tube 14 to outlet 18 and through check valve 108 into mixing chamber 102. The liquid and gas are forced together in mixing chamber 102 to form a gas-infused liquid that passes out of mixing chamber 102 through outlet 109. In some embodiments, the gas-infused liquid is a beverage or used in the making of a beverage.

Capillary tube 14 includes a coil 26 and stems 22, 22 extending from opposing ends of coil 26 as shown in FIGS. 2 and 3. A circuitous path 28 is formed through coil 26 and stems 22, 24 of capillary tube 14. Fitting 12 is formed to include openings 32, 34. Opening 32 extends into a cavity 36 of inlet 16 and opening 34 extends into a cavity 38 of outlet 18 as shown in FIG. 3. Cavities 36, 38 are separated from one another in fitting 12. Stem 22 of capillary tube 14 is positioned in opening 32 of fitting 12 and stem 24 is positioned in opening 34 as shown in FIG. 3. Gas is supplied into cavity 36 of inlet 16 and passes through circuitous path 28 of capillary tube 14 to cavity 38 of outlet 18. In some embodiments, a filter 40 is coupled to inlet 16 as shown in FIG. 3. Filter 40 is porous to allow the gas to pass through filter 40.

Capillary tube 14 controls a volume of gas delivered from cavity 36 to cavity 38 substantially independent of pressure fluctuations of the gas in cavity 36. The relatively small diameter and long length of circuitous path 28 regulates the amount of gas that can pass from cavity 36 to cavity 38. The diameter and length of circuitous path 28 can be adjusted to adjust the volume (or flow rate) of gas that passes through gas-volume controller 10. In some embodiments, one gas-volume controller 10 can be exchanged for another gas-volume controller 10 having a different configuration to allow a different volume of gas to be supplied to mixing chamber 102. In some embodiments, one gas-volume controller 10 can be exchanged for another gas-volume controller 10 of the same configuration for replacement.

Gas-volume controller 10 provides an inexpensive way to regulate gas volume delivered to mixing chamber 102. Gas-volume controller 10 is easily replaceable and exchangeable. Regulating gas flow into mixing chamber 102 maximizes consistency of the gas-infused liquid being formed by gas infuser 100. In some embodiments, no check valves are used in gas infuser 100, and inlet 101 is formed on injector 104 and gas-volume controller 10 is coupled directly to mixing chamber 102. In some embodiments, a liquid-volume controller similar to gas-volume controller 10 is used to regulate liquid volume delivered to mixing chamber 102 through inlet 101. In some embodiments, a gas-infused liquid-volume controller similar to gas-volume controller 10 is used to regulate gas-infused liquid volume exiting mixing chamber 102 through outlet 109.

6

Another embodiment of a gas infuser 200 in accordance with the present disclosure is shown in FIG. 4. Gas infuser 200 includes a mixing chamber 202 and a liquid injector 204 coupled to mixing chamber 202. A check valve 206 is coupled to injector 204 and provides an inlet 201 for passing a liquid through injector 204 into mixing chamber 202. In the illustrative embodiment, a gas-volume controller 210 in accordance with the present disclosure is coupled to mixing chamber 202 to control a volume of gas delivered to mixing chamber 202. Gas-volume controller 210 includes a fitting 212 and a plug 214 coupled to fitting 212 as shown in FIG. 5. Fitting 212 has an inlet 216 and an outlet 218, and outlet 218 is coupled to check valve 208. Plug 214 is received in a cavity 222 of fitting 212 and engages with threads 224 to form a circuitous path between inlet 216 to outlet 218 as suggested in FIGS. 6-8.

Injector 204 is received in mixing chamber 202 and includes a plurality of slots 203 to allow liquid to pass through injector 204 into mixing chamber 202 as suggested in FIG. 6. A liquid, such as water or pre-formed beverage for example, passes from inlet 201 through check valve 206 and injector 204 into mixing chamber 202 as suggested in FIG. 4. A gas, such as CO₂ or N₂, passes from inlet 216 through the circuitous path defined between fitting 212 and plug 214 to outlet 218 and through check valve 208 into mixing chamber 202. The liquid and gas are forced together in mixing chamber 202 to form a gas-infused liquid that passes out of mixing chamber 202 through outlet 209. In some embodiments, the gas-infused liquid is a beverage or used in the making of a beverage.

Plug 214 includes a body 232 and an exterior surface 234 as shown in FIG. 7. Plug 214 is sized such that exterior surface 234 engages with threads 224 when plug 214 is received in cavity 222 of fitting 212. In the illustrative embodiment, threads 224 are helical. Plug 214 is pressed into cavity 222 along an axis A. Body 232 separates inlet 216 from outlet 218 when received in cavity 222. A circuitous path 240 is formed along threads 224 between fitting 212 and plug 214 as suggested in FIG. 8. Threads 224 engage with exterior surface 234 to form a seal such that adjacent portions of the circuitous path 240 are separated from one another to force the gas supplied to inlet 216 to pass along circuitous path 240 to outlet 218. In some embodiments, a filter 219 is coupled to inlet 216 as shown in FIG. 6. Filter 219 is porous to allow the gas to pass through filter 219.

Circuitous path 240 controls a volume of gas delivered from inlet 216 to outlet 218 substantially independent of pressure fluctuations of the gas at inlet 216. The relatively small diameter and long length of circuitous path 240 regulates the amount of gas that can pass from inlet 216 to outlet 218. The diameter and length of circuitous path 240 can be adjusted to adjust the volume (or flow rate) of gas that passes through gas-volume controller 210. In some embodiments, one gas-volume controller 210 can be exchanged for another gas-volume controller 210 having a different configuration to allow a different volume of gas to be supplied to mixing chamber 202. In some embodiments, one gas-volume controller 210 can be exchanged for another gas-volume controller 210 of the same configuration for replacement.

Gas-volume controller 210 provides an inexpensive way to regulate gas volume delivered to mixing chamber 202. Gas-volume controller 210 is easily replaceable and exchangeable. Regulating gas flow into mixing chamber 202 maximizes consistency of the gas-infused liquid being formed by gas infuser 200. In some embodiments, no check valves are used in gas infuser 200, and inlet 201 is formed

7

on injector **204** and gas-volume controller **210** is coupled directly to mixing chamber **202**. In some embodiments, a liquid-volume controller similar to gas-volume controller **210** is used to regulate liquid volume delivered to mixing chamber **202** through inlet **201**. In some embodiments, a gas-infused liquid-volume controller similar to gas-volume controller **210** is used to regulate gas-infused liquid volume exiting mixing chamber **202** through outlet **209**.

Another embodiment of a gas-volume controller **310** in accordance with the present disclosure is shown in FIG. 9. Gas-volume controller **310** is similar to gas-volume controller **210** shown in FIGS. 4-8, and includes a fitting **312** and a plug **314** coupled to fitting **312**. Fitting **312** has an inlet **316** and an outlet **318**. Plug **314** is received in a cavity **322** of fitting **312** to engage threads **334** with an interior surface **324** to form a circuitous path between inlet **316** to outlet **318** as suggested in FIG. 10.

Plug **314** includes a body **332** and threads **334** extending outward from body **332** as shown in FIG. 9. In the illustrative embodiment, threads **334** are helical. Plug **314** is sized such that threads **334** engage with interior surface **324** when plug **314** is received in cavity **322** of fitting **312**. Plug **314** is pressed into cavity **322** along an axis A. Body **332** separates inlet **316** from outlet **318** when received in cavity **322**. A circuitous path **340** is formed along threads **334** between fitting **312** and plug **314** as suggested in FIG. 10. Threads **334** engage with interior surface **324** to form a seal such that adjacent portions of the circuitous path **340** are separated from one another to force the gas supplied to inlet **316** to pass along circuitous path **340** to outlet **318**. In some embodiments, a liquid-volume controller similar to gas-volume controller **310** is used to regulate liquid volume delivered to mixing chamber **202** through inlet **201**. In some embodiments, a gas-infused liquid-volume controller similar to gas-volume controller **310** is used to regulate gas-infused liquid volume exiting mixing chamber **202** through outlet **209**.

Another embodiment of a gas-volume controller **410** in accordance with the present disclosure is shown in FIG. 11. Gas-volume controller **410** is similar to gas-volume controller **210** shown in FIGS. 4-8, and includes a fitting **412** and a plug **414** coupled to fitting **412**. Fitting **412** has an inlet **416** and an outlet **418**. Plug **414** is received in a cavity **422** of fitting **412** to engage threads **434** of plug **414** with threads **424** of fitting **412** to form a circuitous path between inlet **416** to outlet **418** as suggested in FIG. 12.

Plug **414** includes a body **432** and threads **434** extending outward from body **432** as shown in FIG. 11. In the illustrative embodiment, threads **424**, **434** are helical. Plug **414** is sized such that threads **434** engage with threads **424** of fitting **412** when plug **414** is received in cavity **422**. Plug **414** is rotated to engage threads **424**, **434** to drive plug **414** into cavity **422** along an axis A. Body **432** separates inlet **416** from outlet **418** when received in cavity **422**.

A circuitous path **440** is formed along threads **424**, **434** between fitting **412** and plug **414** as suggested in FIG. 12. In the illustrative embodiment, threads **434** are shorter relative to threads **424** such that threads **434** extend partially into threads **424**. A relative height of threads **424**, **434** can be adjusted to adjust a size of circuitous path **440**. Threads **424**, **434** engage with one another to form a seal such that adjacent portions of circuitous path **440** are separated from one another to force the gas supplied to inlet **416** to pass along circuitous path **440** to outlet **418**. In some embodiments, a liquid-volume controller similar to gas-volume controller **410** is used to regulate liquid volume delivered to mixing chamber **202** through inlet **201**. In some embodi-

8

ments, a gas-infused liquid-volume controller similar to gas-volume controller **410** is used to regulate gas-infused liquid volume exiting mixing chamber **202** through outlet **209**.

Another embodiment of a gas infuser **500** in accordance with the present disclosure is shown in FIG. 13. Gas infuser **500** includes a mixing chamber **502** and a liquid injector **504** coupled to mixing chamber **502**. An inlet **501** is coupled to mixing chamber **502** for passing a liquid through injector **504** into mixing chamber **502**. A check valve **506** is coupled to inlet **501**. In the illustrative embodiment, a gas-volume controller **510** in accordance with the present disclosure is coupled to mixing chamber **502** to control a volume of gas delivered to mixing chamber **502**.

Gas-volume controller **510** includes a fitting **512** and a plug **514** coupled to fitting **512** as suggested in FIGS. 13 and 14. Fitting **512** has an inlet **516** and an outlet **518**, and outlet **518** is coupled to mixing chamber **502**. A check valve **508** is coupled to outlet **518** of fitting **512**. Plug **514** is received in and engages with fitting **512** to form a circuitous path between inlet **516** to outlet **518**.

Injector **504** is received in a cavity **562** of mixing chamber **502** and includes a plurality of slots **503** to allow liquid to pass through injector **504** into mixing chamber **502** as suggested in FIGS. 15 and 16. Inlet **501** is at least partially received in cavity **562** and a clip **552** engages with inlet **501** and mixing chamber **502** to block removal of inlet **501** and injector **504** from cavity **562**. Clip **552** includes a holder **551** for engaging with inlet **501** and a fastener **553** for securing holder **551** to mixing chamber **502**. Fitting **512** of gas-volume controller **510** is at least partially received in a cavity **564** of mixing chamber **502** that intersects with cavity **562** to allow gas flow through mixing chamber **502** to outlet **509**. A clip **554**, similar to clip **552**, blocks removal of fitting **512** from cavity **564**. A connector **542** is received in inlet **516** of fitting **512** and secured in place with a clamp **544**.

A liquid, such as water or pre-formed beverage for example, passes from inlet **501** through check valve **506** and injector **404** into mixing chamber **502** as suggested in FIG. 17. A gas, such as CO₂ or N₂, passes from connector **542** through inlet **516** and the circuitous path defined between fitting **512** and plug **514** to outlet **518**, and through check valve **508**, into mixing chamber **502**. The liquid and gas are forced together in mixing chamber **502** to form a gas-infused liquid that passes out of mixing chamber **502** through outlet **509**. In some embodiments, the gas-infused liquid is a beverage or used in the making of a beverage.

Plug **514** includes a body **532** and a head **534** coupled to body **532** as shown in FIGS. 14 and 18. A passage **536** is formed in plug **514** to allow gas to flow through body **532** and around head **534**. Passage **536** includes a bore **531** extending axially through body **532** and a bore **533** extending through plug **514** between body **532** and head **534**. In some embodiments, a filter **519** is received in bore **531** as shown in FIG. 18. Filter **519** is porous to allow the gas to pass through filter **519**.

A cavity **522** is formed through fitting **512** defining inlet **516** and outlet **518** as suggested in FIGS. 14 and 18. Threads on body **532** of plug **514** engage with internal threads **526** of cavity **522** to position plug **514** relative to fitting **512** as suggested in FIGS. 18 and 19. Threads on head **534** of plug **514** engage with an interior surface **524** of fitting **512** to define the circuitous path. In the illustrative embodiment, a tool slot **538** allows a user to engage with plug **514** to adjust a depth D of head **534** relative to interior surface **524**. Adjustment of depth D adjusts a length of the circuitous path for setting a volume rate of gas that can pass from inlet **516**

to outlet **518**. Body **532** engages with threads **526** to block gas from passing around body **532**. Gas flows from inlet **516** through passage **536** and around head **534** along the circuitous path defined between head **534** and interior surface **524** toward outlet **518**. In the illustrative embodiment, the threads on head **534** are helical.

The circuitous path controls a volume of gas delivered from inlet **516** to outlet **518** substantially independent of pressure fluctuations of the gas at inlet **516**. The relatively small diameter and long length of the circuitous path regulates the amount of gas that can pass from inlet **516** to outlet **518**. The diameter and length of the circuitous path can be adjusted to adjust the volume (or flow rate) of gas that passes through gas-volume controller **510**. In some embodiments, one gas-volume controller **510** can be exchanged for another gas-volume controller **510** having a different configuration to allow a different volume of gas to be supplied to mixing chamber **502**. In some embodiments, one gas-volume controller **510** can be exchanged for another gas-volume controller **510** of the same configuration for replacement.

Gas-volume controller **510** provides an inexpensive way to regulate gas volume delivered to mixing chamber **502**. Gas-volume controller **510** is easily replaceable and exchangeable. Regulating gas flow into mixing chamber **502** maximizes consistency of the gas-infused liquid being formed by gas infuser **500**. In some embodiments, no check valves are used in gas infuser **500**. In some embodiments, a liquid-volume controller similar to gas-volume controller **510** is used to regulate liquid volume delivered to mixing chamber **502** through inlet **501**. In some embodiments, a gas-infused liquid-volume controller similar to gas-volume controller **510** is used to regulate gas-infused liquid volume exiting mixing chamber **502** through outlet **509**.

An embodiment of a check valve **508** in accordance with the present disclosure is shown in FIGS. **20** and **21**. Check valve **508** includes a frame **572** and a plunger **574** coupled to frame **572**. Plunger **574** is biased by a spring **576** toward a hole **578** formed through frame **572**. Plunger **574** is movable relative to frame **572** toward and away from hole **578**. A gasket **575** carried on plunger **574** seals against hole **578** in a closed position. A gasket **573** seals against a mating component of check valve **508**, such as fitting **512**. Flexible grips **571** formed on frame **572** position check valve **508** relative to the mating component. In some embodiments, check valve **506** is formed similar to check valve **508**.

Another embodiment of a gas infuser **600** in accordance with the present disclosure is shown in FIGS. **22** and **23**. Gas infuser **600** includes a mixing chamber **602** and injectors **604**, **686** coupled to mixing chamber **602**. A liquid inlet **601** is coupled to mixing chamber **602** for passing a liquid through injector **686** into mixing chamber **602**. A check valve **606** is coupled to liquid inlet **601**. An additive inlet **682** is coupled to mixing chamber **602** for passing an additive into mixing chamber **602** for combination with the liquid to form a liquid-additive solution. A check valve **681** is coupled to additive inlet **682**. The liquid-additive solution passes through injector **604** for mixing with a gas in mixing chamber **602**. In the illustrative embodiment, a gas-volume controller **610** in accordance with the present disclosure is coupled to mixing chamber **602** to control a volume of gas delivered to mixing chamber **602**.

Gas-volume controller **610** is similar to gas-volume controller **510** shown in FIGS. **13-19**, and includes a fitting **612** and a plug **614** coupled to fitting **612** as suggested in FIGS. **22** and **24**. Fitting **612** has an inlet **616** and an outlet **618**, and outlet **618** is coupled to mixing chamber **602**. A check valve **608** is coupled to outlet **618** of fitting **612**. Plug **614** is

received in and engages with fitting **612** to form a circuitous path between inlet **616** to outlet **618**.

Injector **686** is received in a cavity **662** of mixing chamber **602** and includes a plurality of slots **683** to allow liquid to pass through injector **686** into mixing chamber **602** as suggested in FIGS. **24** and **25**. Liquid inlet **601** is at least partially received in cavity **662** and a clip **652**, similar to clip **552** shown in FIG. **15**, engages with liquid inlet **601** and mixing chamber **602** to block removal of liquid inlet **601** and injector **686** from cavity **662**. Additive inlet **682** is at least partially received in a cavity **667** of mixing chamber **602** and a clip **658**, similar to clip **552** shown in FIG. **15**, engages with additive inlet **682** and mixing chamber **602** to block removal of additive inlet **682** from cavity **667**. Cavity **667** is connected to cavity **662** by a cavity **666**. A flow adjuster **684** extends into cavity **666** to control an amount of additive flowing into cavity **662** for mixing with the liquid.

Cavity **662** of mixing chamber **602** extends into a cavity **668** to allow passage of the liquid-additive solution toward outlet **609** as suggested in FIGS. **24** and **25**. Injector **604** is received in cavity **668** and includes a plurality of slots **603** to allow the liquid-additive solution to pass through injector **604**. Outlet **609** is at least partially received in cavity **668** and a clip **656**, similar to clip **552** shown in FIG. **15**, engages with outlet **609** and mixing chamber **602** to block removal of outlet **609** and injector **604** from cavity **668**. In the illustrative embodiment, a spacer **690** including two interlocking plates **692**, **694** is received in outlet **609** and configured to engage with injector **604** to block movement of injector **604** toward outlet **609**. Fitting **612** of gas-volume controller **610** is at least partially received in a cavity **664** of mixing chamber **602** that intersects with cavity **668** to allow gas flow through mixing chamber **602** to outlet **609**. A clip **654**, similar to clip **552** shown in FIG. **15**, blocks removal of fitting **612** from cavity **664**.

A liquid, such as water or pre-formed beverage for example, passes from liquid inlet **601** through check valve **606** and injector **686** into mixing chamber **602** as suggested in FIG. **26**. The additive, such as a concentrate, flavoring, or other substance, passes from additive inlet **682** through check valve **681** into mixing chamber **602**. The additive flows through cavity **666** and is forced together with the liquid to form a liquid-additive solution. The liquid-additive solution flows through injector **604** into cavity **668**. A gas, such as CO₂ or N₂, passes from inlet **616** of gas-volume controller **610** through the circuitous path defined between fitting **612** and plug **614** to outlet **618**, and through check valve **608**, into mixing chamber **602**. The liquid-additive solution and gas are forced together in mixing chamber **602** to form a gas-infused liquid that passes out of mixing chamber **602** through outlet **609**. In some embodiments, the gas-infused liquid is a beverage or used in the making of a beverage.

The circuitous path controls a volume of gas delivered from inlet **616** to outlet **618** substantially independent of pressure fluctuations of the gas at inlet **616**. The relatively small diameter and long length of the circuitous path regulates the amount of gas that can pass from inlet **616** to outlet **618**. The diameter and length of the circuitous path can be adjusted to adjust the volume (or flow rate) of gas that passes through gas-volume controller **610**. In some embodiments, one gas-volume controller **610** can be exchanged for another gas-volume controller **610** having a different configuration to allow a different volume of gas to be supplied to mixing chamber **602**. In some embodiments, one gas-volume controller **610** can be exchanged for another gas-volume controller **610** of the same configuration for replacement.

11

Gas-volume controller **610** provides an inexpensive way to regulate gas volume delivered to mixing chamber **602**. Gas-volume controller **610** is easily replaceable and exchangeable. Regulating gas flow into mixing chamber **602** maximizes consistency of the gas-infused liquid being formed by gas infuser **600**. In some embodiments, no check valves are used in gas infuser **600**. In some embodiments, a liquid-volume controller similar to gas-volume controller **610** is used to regulate liquid volume delivered to mixing chamber **602** through inlet **601**. In some embodiments, an additive-volume controller similar to gas-volume controller **610** is used in combination with or alternatively to flow adjuster **684** to regulate additive volume delivered to mixing chamber **602** through inlet **682**. In some embodiments, a gas-infused liquid-volume controller similar to gas-volume controller **610** is used to regulate gas-infused liquid volume exiting mixing chamber **602** through outlet **609**.

In illustrative embodiments, gas-volume controllers in accordance with the present disclosure regulate the volume of gas provided for mixing with a volume of liquid. The flow rate of the gas does not fluctuate with fluctuations in gas pressure at an inlet of the gas-volume controller to provide consistency in the amount of gas infused with the liquid. In some embodiments, a gas-to-liquid volume ratio is maintained substantially constant with the gas-volume controller regardless of fluctuations in gas pressure at the inlet of the gas-volume controller. In some embodiments, the flow rate of gas supplied to the mixing chamber is maintained substantially constant with the gas-volume controller regardless of fluctuations in gas pressure at the inlet of the gas-volume controller.

In illustrative embodiments, a gas infuser in accordance with the present disclosure for forming a gas-infused liquid includes a mixing chamber, a liquid injector, and a gas-volume controller. The liquid injector is coupled to the mixing chamber and adapted to pass a liquid into the mixing chamber. The gas-volume controller is coupled to the mixing chamber and configured to pass a gas into the mixing chamber for mixing with the liquid to form the gas-infused liquid. The gas-volume controller includes an inlet and an outlet separated from the inlet. The inlet is operatively coupled to a gas supply for receiving the gas. The outlet is operatively coupled to the mixing chamber to pass the gas into the mixing chamber. The gas-volume controller is configured to provide a flow rate of gas exiting the outlet for a set average pressure of the gas at the inlet. The gas-volume controller is formed to define a circuitous path between the inlet and the outlet. The circuitous path is configured to regulate the flow rate of the gas exiting the outlet to be substantially constant such that the flow rate is substantially independent of pressure fluctuations from the set average pressure of the gas at the inlet.

In illustrative embodiments, a gas-volume controller in accordance with the present disclosure includes an inlet, an outlet separated from the inlet, and a circuitous path formed between the inlet and the outlet. The circuitous path is configured to regulate a flow rate of gas exiting the outlet to be substantially constant such that the flow rate is substantially independent of pressure fluctuations from a set average pressure of the gas at the inlet.

While the present disclosure describes various exemplary embodiments, the disclosure is not so limited. To the contrary, the disclosure is intended to cover various modifications, uses, adaptations, and equivalent arrangements based on the principles disclosed. Further, this application is intended to cover such departures from the present disclosure as come within at least the known or customary practice

12

within the art to which it pertains. It is envisioned that those skilled in the art may devise various modifications and equivalent structures and functions without departing from the spirit and scope of the disclosure as recited in the following claims. The scope of the following claims is to be accorded the broadest interpretation to encompass all such modifications and equivalent structures and functions.

What is claimed is:

1. A gas infuser for forming a gas-infused liquid, the gas infuser comprising:

- a mixing chamber;
- a liquid injector coupled to the mixing chamber and adapted to pass a liquid into the mixing chamber; and
- a gas-volume controller coupled to the mixing chamber and configured to pass a gas into the mixing chamber for mixing with the liquid to form the gas-infused liquid, the gas-volume controller including a fitting defining a cavity between an inlet and an outlet spaced apart from the inlet and a plug arranged in the cavity between the inlet and the outlet, the outlet operatively coupled to the mixing chamber to pass the gas into the mixing chamber,

wherein at least one of an interior surface of the fitting and an exterior surface of the plug includes threads, a circuitous helical path is formed along the threads between the fitting and the plug to fluidly couple the inlet to the outlet, the gas-volume controller is configured to provide a flow rate of gas exiting the outlet for a set average pressure of the gas at the inlet, and the circuitous helical path is configured to regulate the flow rate of the gas exiting the outlet to be substantially constant such that the flow rate is substantially independent of pressure fluctuations from the set average pressure of the gas at the inlet.

2. The gas infuser of claim 1, further comprising a filter element coupled to the inlet.

3. The gas infuser of claim 1, further comprising a check valve coupled between the gas-volume controller and the mixing chamber, the check valve configured to block fluid from passing out of the mixing chamber into the gas-volume controller.

4. The gas infuser of claim 1, further comprising a check valve coupled to the liquid injector, the check valve configured to block fluid from passing out of the mixing chamber through the liquid injector.

5. A gas-volume controller comprising:

- a fitting defining a cavity between an inlet and an outlet spaced apart from the inlet; and
- a plug arranged in the cavity between the inlet and the outlet,

wherein at least one of an interior surface of the fitting and an exterior surface of the plug includes threads, wherein a circuitous helical path is formed along the threads between the fitting and the plug to fluidly couple the inlet to the outlet, and wherein the circuitous helical path is configured to regulate a flow rate of gas exiting the outlet to be substantially constant such that the flow rate is substantially independent of pressure fluctuations from a set average pressure of the gas at the inlet.

6. The gas-volume controller of claim 5, further comprising a filter element coupled to the inlet.

7. The gas-volume controller of claim 5, further comprising a check valve coupled to the outlet, the check valve configured to block fluid from passing through the check valve into the gas-volume controller.

13

8. A gas infuser for forming a gas-infused liquid, the gas infuser comprising:

a mixing chamber;

a liquid injector coupled to the mixing chamber and adapted to pass a liquid into the mixing chamber; and

a gas-volume controller coupled to the mixing chamber and configured to pass a gas into the mixing chamber for mixing with the liquid to form the gas-infused liquid, the gas-volume controller including a fitting defining a cavity between an inlet and an outlet separated from the inlet and a plug arranged in the cavity between the inlet and the outlet, the inlet operatively coupled to a gas supply for receiving the gas, the outlet operatively coupled to the mixing chamber to pass the gas into the mixing chamber, and

wherein the plug includes a body and a head coupled to the body, a passage is formed in the plug and configured to allow gas to flow into the body and out of the plug at a position between the body and the head, threads extend along an exterior surface of the head, a circuitous helical path is formed along the threads between the fitting and the head to fluidly couple the inlet to the outlet, the gas-volume controller is configured to provide a flow rate of gas exiting the outlet for a set average pressure of the gas at the inlet, and the circuitous path is configured to regulate the flow rate of the gas exiting the outlet to be substantially constant such that the flow rate is substantially independent of pressure fluctuations from the set average pressure of the gas at the inlet.

14

9. The gas infuser of claim 8, wherein the passage of the plug includes an axial bore extending into the body and a cross bore extending through the plug between the body and the head and in fluid communication with the axial bore.

10. A gas-volume controller comprising:

a fitting defining a cavity between an inlet and an outlet spaced apart from the inlet; and

a plug arranged in the cavity between the inlet and the outlet,

wherein the plug includes a body and a head coupled to the body, a passage is formed in the plug and configured to allow gas to flow into the body and out of the plug at a position between the body and the head, and threads extend along an exterior surface of the head, wherein a circuitous path is formed along the threads between the fitting and the head to fluidly couple the inlet to the outlet, and

wherein the circuitous path is configured to regulate a flow rate of gas exiting the outlet to be substantially constant such that the flow rate is substantially independent of pressure fluctuations from a set average pressure of the gas at the inlet.

11. The gas-volume controller of claim 10, wherein the passage of the plug includes an axial bore extending into the body and a cross bore extending through the plug between the body and the head and in fluid communication with the axial bore.

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